

Mediterranean Diet in Predementia and Dementia Syndromes

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Abstract: There is a critical need to potentially individualize new strategies able to prevent and to slow down the progression of predementia and dementia syndromes. Only recently higher adherence to a Mediterranean-type diet was associated with decreased cognitive decline although the Mediterranean diet (MeDi) combines several foods, micro- and macronutrients already separately proposed as potential protective factors against dementia and predementia syndromes. In fact, elevated saturated fatty acids could have negative effects on age-related cognitive decline and mild cognitive impairment (MCI). Furthermore, at present, epidemiological evidence suggested a possible association among fish consumption, monounsaturated fatty acids and polyunsaturated fatty acids (PUFA) (particularly, n-3 PUFA) and reduced risk of cognitive decline and dementia. Light to moderate alcohol use may be associated with a reduced risk of incident dementia and Alzheimer's disease (AD), while for vascular dementia, cognitive decline, and predementia syndromes the current evidence is only suggestive of a protective effect. Finally, the limited epidemiological evidence available on fruit and vegetable consumption and cognition generally supported a protective role of these macronutrients against cognitive decline, dementia, and AD. Moreover, recent prospective studies provided evidence that higher adherence to a Mediterranean-type diet could be associated with slower cognitive decline, reduced risk of progression from MCI to AD, reduced risk of AD, and decreased all-causes mortality in AD patients. These findings suggested that adherence to the MeDi may affect not only the risk for AD, but also for predementia syndromes and their progression to overt dementia. Nonetheless, at present, no definitive dietary recommendations are possible. However, high levels of consumption of fats from fish, vegetable oils, non-starchy vegetables, low glycemic fruits, and diet low in foods with added sugars and with moderate wine intake should be encouraged. In fact, this dietary advice is in accordance with recommendations for lowering the risk of cardiovascular disease, obesity, diabetes, and hypertension and might open new ways for the prevention and management of cognitive decline and dementia.

Keywords: Mediterranean diet adherence, vegetables, fruits, alcohol consumption, MUFA, PUFA, predementia syndromes, dementia, Alzheimer's disease, vascular dementia, mild cognitive impairment, age-related cognitive decline.

INTRODUCTION

The prevalence and incidence of dementia increase with age particularly in the very elderly, over the ages of 75 and 80 [1]. Alzheimer's disease (AD) is the most common form of dementia, followed by vascular dementia (VaD) [2]. AD is characterized by relatively slow chronic but progressive neurodegeneration and impairment in cognition, behavior, and functionality. The disease is both multifactorial and heterogeneous, involving aberrant protein processing and is characterized by the presence of both intraneuronal protein clusters composed of paired helical filaments of hyperphosphorylated tau protein [neurofibrillary tangles (NFTs)], and extracellular protein aggregates [senile plaques (SPs)]. The SPs are the result of misprocessing of the amyloid precursor protein (APP), a type-1 transmembrane protein, by β - and γ -secretases to form a toxic β -amyloid (A β) peptide of 40–42

amino acids [3] that aggregates and initiates a pathogenic self-perpetuating cascade ultimately leading to neuronal loss and dementia. According to the "amyloid cascade hypothesis" [4], the development of SPs is thought to precede and precipitate the formation of NFTs as a result of the cellular changes invoked, and the oligomeric forms of A β peptide is the main cause of neuronal death in AD.

The term "predementia syndrome" identified all conditions with age-related deficits in cognitive function reported in the literature, including a mild stage of cognitive impairment based on a normality model and pathological conditions considered predictive or early stages of dementia [5]. Therefore, this "umbrella term" includes different conditions and, among them, MCI is at present the most widely used term to indicate non-demented aged persons with no significant disability and a mild memory or cognitive impairment which cannot be explained by any recognized medical or psychiatric condition [6]. There is now ample evidence that MCI is often a pathology-based condition with a high rate of progression to AD [7]. Therefore, MCI has also been identi-

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fied as the predementia syndrome for AD, while other predementia syndromes have not yet been operationalized for other specific forms of dementia. The more recently proposed multiple subtypes of MCI were intended to reflect the heterogeneity of different types of dementia. Actually, there are at least three different subclassifications of MCI according to its cognitive features, clinical presentation, and probable etiology [7]. MCI definitions can be broadly classified as amnesic (aMCI) and nonamnesic (naMCI). Therefore, aMCI, with a central role for memory disorder and with relative preservation of other cognitive domains, can be subdivided into a single domain subtype with a pronounced memory deficit or a multiple domain subtype that includes memory impairment along with some impairment in other cognitive domains such as language, executive function, and visuospatial skills [6, 7]. The other major MCI subtype is naMCI, which similarly can be subdivided into single and multiple domain subtypes [6, 7]. Furthermore, different diagnostic criteria have been proposed for other predementia syndromes, and the terms age-related cognitive decline (ARCD) [8] and aging-associated cognitive decline (AACD) [9] have been recently proposed to identify individuals with mild cognitive disorders associated with aging, also non pathological-based and without progression to dementia. In particular, ARCD is defined by the DSM-IV as "an objectively identified decline in cognitive functioning consequent to the aging process that is within normal limits given the person's age", but there are no defined diagnostic criteria, and few epidemiological studies using this definition have been conducted [5].

In the last decade, advances in understanding the neurobiology of AD have translated into an increase in clinical trials assessing various potential AD treatments [10]. However, only cholinesterase inhibitors (ChEIs) and the N-methyl-D-aspartic acid (NMDA)-receptor antagonist memantine have received Food and Drug Administration (FDA) approval for the symptomatic treatment of AD [10].

Epidemiological studies mainly with cross-sectional data have suggested that non-steroidal anti-inflammatory drugs (NSAIDs), estrogens, HMG-CoA reductase inhibitors (statins), and tocopherol (vitamin E) may be beneficial to reduce the incidence of dementia and AD [10]. However, bias of case selection and several other sources of error are inherent in epidemiological studies and subsequent clinical trials have often been disappointing.

The causes of dementia and predementia syndromes are, at present, unknown, but some studies have suggested that it may be prevented [11-14]. In fact, epidemiological evidence supported the hypothesis that modifiable vascular and lifestyle-related factors were associated to the development of dementia and predementia syndromes in late life, opening new avenues for the prevention of these diseases [11, 13]. The vascular and related factors that have been associated with dementia and cognitive decline included hypertension and elevated blood pressure [15], total cholesterol [16], diabetes mellitus [17], body mass index (BMI) [17], and the metabolic syndrome [18, 19]. Also the presence of cardiovascular and other chronic diseases [20], depression [7], and low levels of physical activity [13] have been identified as risk factors for cognitive decline. In contrast, physical exer-

cise [13, 21] and education [13, 22] appeared to have clear protective effects on cognitive functioning and possible reduction in incident dementia. For the latter, whether this is due to education per se or related to lifetime occupation and or level of cognitive exercise is debated [22].

Furthermore, deficiencies of some micronutrients (especially, vitamins B1, B2, B6, B12, C, and folate) have been described quite frequently in elderly people and found to be significantly associated with cognitive impairment [23]. However, the role of the diet in cognitive decline has not been extensively investigated, with a few data available on the role of macronutrient intake in the pathogenesis of predementia and dementia syndromes [12, 13, 23-29]. Since several dietary factors affect the risk of cardiovascular disease, it can be assumed that they also influence the risk of dementia [30]. This concept is further supported by recent evidence that certain diets have been associated with a lower incidence of AD. In fact, recent findings demonstrated that maintaining a healthy diet may impact on many of these possible risk factors for cognitive decline and the model to follow seems to be the Mediterranean diet (MeDi) [12]. The typical dietary pattern of MeDi is characterized by high intakes of vegetables, fruits and nuts, legumes, cereals, fish, and monounsaturated fatty acids (MUFA); relatively low intakes of meat, and dairy products, and moderate consumption of alcohol. In fact, higher levels of consumption of olive oil, very rich in MUFA, are considered the hallmark of the traditional MeDi, and some recent studies have suggested that dietary fatty acids, particularly high MUFA intake, regular fish and polyunsaturated fatty acid (PUFA) n-3 consumption, may play a role in the prevention of cognitive decline associated with aging or dementia [31-33]. Elevated dietary MUFA and PUFA n-3 and high fish consumption alongside with high levels of antioxidants from fruit and vegetables [34, 35], and moderate alcohol consumption [36, 37] may permit a beneficial effect on the risk of dementia.

In this review article, we examined the possible role of macronutrients and food nutrients of the Mediterranean diet and the adherence to the Mediterranean dietary pattern in modulating the risk of age-related changes in cognitive function, predementia syndromes, and dementia as well as the possible mechanisms behind the observed associations from the English literature published before December 2009. We reviewed clinical and epidemiological studies from the international literature, including both cross-sectional and longitudinal studies that provided a description of the diagnostic criteria used for predementia or dementia syndromes. We searched through the PubMed database of NCBI (available at <http://www.ncbi.nlm.nih.gov>) by author and the following keywords: Mediterranean, diet, Mediterranean dietary pattern, adherence, monounsaturated fatty acids, polyunsaturated fatty acids, alcohol consumption, cereals, fruits, vegetables, mild cognitive impairment, dementia, Alzheimer's disease, vascular dementia, predementia syndromes, and MCI.

MACRONUTRIENTS FROM MEDITERRANEAN DIET AND COGNITIVE DECLINE

The MeDi, in the south of Europe, represented the dietary pattern usually consumed among the populations bordering

the Mediterranean sea, and it has been widely reported to be a model of healthy eating for its contribution to a favourable health status and a better quality of life [38]. Since the first data of Ancel Keys from the Seven Countries' Study in the 1950s to 1960s [39], several studies in different populations have established a beneficial role for the main components of the MeDi on the occurrence of cardiovascular diseases and chronic degenerative diseases [40, 41]. The Mediterranean-style diet was characterized by abundant plant foods consumption in the form of fruits, vegetables, breads, other forms of cereals, potatoes, beans, nuts and seeds; fresh fruit as the typical dessert; olive oil as the main source of MUFA; dairy products as principally cheese and yogurt; a low-to-moderate consumption of fish depending on the proximity of the sea; a low-to moderate consumption of poultry; fewer than four eggs consumed per week; low amount of red meat and wine consumed in low-to-moderate amounts, normally during meals [42]. However, there is no single MeDi but several definitions, because dietary habits vary considerably across the Mediterranean countries bordering the sea [43]. Previous observational studies already indicated that specific foods or nutrients that take part in the traditional MeDi (i.e., fish, unsaturated fatty acids, antioxidant as vitamin E, vitamin B12, folates, carotenes, flavonoids and moderate alcohol) may have potential protective effect against dementia or cognitive decline [12, 44]. In the last years, the study approach was to associate single micro- or macronutrients to ARCD, MCI, AD, or VaD. In this picture, several hallmarks of the MeDi were linked to increased risk or with a protective effect against cognitive impairment [12].

Dietary Fatty Acids, Fish Consumption, and Cognitive Decline

Fatty acids can be categorized briefly into saturated fatty acids (SFA) and unsaturated fatty acids (UFA). SFA, such as stearic acid, are present in products such as meat, dairy products, cookies and pastries. MUFA are most frequently consumed in olive oil. The principal series of PUFA are n-6 (i.e. linoleic acid) and n-3 [i.e., α -linolenic acid, docosahexaenoic acid (DHA), and eicosapentaenoic acid (EPA)]. In our Mediterranean dietary pattern the main sources of n-6 PUFA are vegetable oils, while the principal sources of n-3 PUFA are fatty fish (salmon, tuna, and mackerel). In fact, olive oil contains 70–80% MUFA (oleic acid) and 8–10% PUFA (6–7% linoleic acid and 1–2% α -linolenic acid) [31, 33].

At present, an increasing number of epidemiological and clinical studies have addressed the link between UFA intake and cognitive function, most being cross-sectional [31, 33, 45–48]. The cross-sectional association between dietary macronutrients and cognitive impairment was examined in elderly subjects aged 65–84 years from the Italian Longitudinal Study on Aging (ILSA). After adjustment for educational level, the odds ratios of cognitive decline [Mini-Mental State Examination (MMSE) score < 24] decreased exponentially with the increase of MUFA energy intakes. Furthermore, selective attention performances were independently associated with MUFA intake [45]. An association between fatty acids and cognitive functioning has also been verified in several longitudinal epidemiological studies [31, 49–58] (Table 1), indicating a crucial need for prospective studies that

could confirm initial cross-sectional observations. Therefore, on the basis of the previous significant suggestions [31], the hypothesis that high MUFA and PUFA intakes may protect against the development of cognitive impairment over time in a median follow-up of 8.5 years of the ILSA was further tested. The major finding of this study was that high MUFA, PUFA, and total energy intake were significantly associated with a better cognitive performance in time. No other individual dietary components of this study population was significantly associated with cognitive impairment in time [54]. Finally, other epidemiological findings from the ILSA demonstrated that while dietary fatty acids intakes were not associated with incident MCI, high PUFA intake appeared to have borderline non-significant trend for a protective effect against the development of MCI [55]. Future studies on this topic will address different constructs of predementia syndromes and subtypes of MCI, also accounting for the possible role of apolipoprotein E (APOE) $\epsilon 4$ allele in modifying the possible associations between dietary fatty acids and cognitive decline [33].

As seen above, MUFA, consequently to the high consumption of extra-virgin olive oil, represented the most important fat in MeDi. In North American diets, a major source of MUFA is canola oil, while olive oil consumption is much less frequent. Cumulative evidence suggests that extra-virgin olive oil may have a role in the protection against cognitive decline, other than against coronary artery disease (CAD) and several types of cancer because of its high levels MUFA and polyphenolic compounds [31, 33]. Very recently, in the Three-City (3C) Study, olive oil consumption habits were significantly associated with selective cognitive deficit and cognitive decline, independently of other dietary intakes and after adjusting for potential confounders. Intensive use of olive oil in diet is associated with lower odds of cognitive deficit in visual memory and verbal fluency even if the relationship is not significant when assessing global cognitive functioning with the MMSE [59]. In the analyses of 732 participants of the European Prospective Investigation into Cancer and Nutrition (EPIC) Greeks cohort, intake of olive oil, MUFA and SFA exhibited a weak but not significant positive association with cognitive function as assessed by the MMSE, six to 13 years after enrolment [56].

The possible relationship between dietary fatty acids intake and risk of dementia and AD has been also evaluated in a series of longitudinal studies with contrasting findings [60–67] (Table 1). However, the sum of evidence suggested that an increase of SFA could have negative effects on cognitive functions and incident dementia, while a clear reduction of risk for cognitive decline has been found in population samples with elevated fish consumption, high intake of MUFA and PUFA, particularly n-3 PUFA [33]. In fact, fish, particularly fatty fish (e.g., herring, mackerel, salmon, or trout), is the principal source of n-3 PUFA in the MeDi. Very recently, the baseline data from the Older People And n-3 Long-chain polyunsaturated fatty acid (OPAL) study suggested that higher fish consumption is associated with better cognitive function in later life [68]. Longitudinal observational studies reporting associations of fish consumption with cognitive function have shown mixed results; some longitudinal studies have reported a positive association with higher fish consumption [57, 69–71], while others

Table 1. Principal Longitudinal Clinical and Epidemiological Studies on the Relationships among Dietary Fatty Acids, Fish Consumptions and Dementia (i.e., Alzheimer' Disease, AD, and Vascular Dementia, VaD) or Predementia Syndromes (i.e., Age-Related Cognitive Decline, ARCD, and Mild Cognitive Impairment, MCI)

Reference	Setting and Study Design	Subjects	Dietary Assessment	Cognitive Outcomes	Results and Conclusions
Dietary Fatty Acids, Fish Consumption and Predementia Syndromes					
Kalmijn <i>et al.</i> , 1997 [49] The Zutphen Elderly Study The Netherlands	Longitudinal, population-based (3 years)	476 subjects, aged 69-89 years	Evaluation of dietary intake with the cross-check dietary history method	Cognitive impairment defined as a MMSE score <25 points and cognitive decline as a drop of >2 points of MMSE over a 3-year period	High linoleic acid intake (PUFA) was positively associated with cognitive impairment. High fish consumption was inversely associated with cognitive impairment
Morris <i>et al.</i> , 2004 [51] The Chicago Health and Aging Project (CHAP) USA	Longitudinal, population-based (6 years)	2,560 subjects, aged 65 years and older	Evaluation of dietary intake with a 139-items FFQ	Cognitive change at 3-year and 6-year follow-ups measured with the EBMT of Immediate and Delayed Recall, the MMSE, and the SDMT	A diet high in saturated and <i>trans</i> -unsaturated fat or low in nonhydrogenated unsaturated fats may be associated with cognitive decline among older people
Morris <i>et al.</i> , 2005 [65] Chicago Health and Aging Project (CHAP) USA	Longitudinal, population-based (6 years)	3,718 subjects, aged 65 years and older	Evaluation of dietary intake with a 139-items FFQ	Cognitive change at 3-year and 6-year follow-ups measured with the EBMT of Immediate and Delayed Recall, the MMSE, and the SDMT	Dietary intake of fish was inversely associated with cognitive decline over 6 years. There were no consistent associations with the n-3 fatty acids, although the effect estimates were in the direction of slower decline
Solfrizzi <i>et al.</i> , 2006 [54] The Italian Longitudinal Study on Aging (ILSA) Italy	Longitudinal, population-based (8.5 years)	278 subjects, 65-84 years old from a cohort of 5,632 subjects	Evaluation of MUFA and PUFA dietary intakes with a 77-item FFQ	MMSE	High MUFA, PUFA, and total energy intake were significantly associated with a better cognitive performance in time. The association between high MUFA, PUFA intakes and cognitive performance remained robust even after adjustment for potential confounding variables such as age, sex, educational level, CCI, BMI, and total energy intakes
Solfrizzi <i>et al.</i> , 2006 [55] The Italian Longitudinal Study on Aging (ILSA) Italy	Longitudinal, population-based (2.6 years)	278 subjects, 65-84 years old from a cohort of 5,632 subjects	Evaluation of MUFA and PUFA dietary intakes with a 77-item FFQ	Incident MCI. Diagnostic criteria for MCI: 1.5 SDs below mean age- and education-adjusted on the MMSE and 10th percentile below age- and education-adjusted on memory test, without SMC and intact ADL/IADL	Dietary fatty acids intakes were not associated with incident MCI. However, high PUFA intake appeared to have borderline non-significant trend for a protective effect against the development of MCI that may be important.
Psaltopoulou <i>et al.</i> , 2008 [56] European Prospective Investigation into Cancer and Nutrition (EPIC) Greece	Longitudinal, population-based (median 8 years)	732 subjects, 60 years or older	Evaluation of dietary intakes with a 150-item FFQ. A dietary composite score (MeDi score) evaluated adherence to Mediterranean diet	MMSE	No significant association between MeDi score and MMSE scores, whereas a statistically significant inverse association was found between MMSE performance and some individual dietary components, such as seed oil or PUFA intakes
Eskilinen <i>et al.</i> , 2008 [57] Cardiovascular Risk Factors Aging and Dementia (CAIDE) Finland	Longitudinal, population-based (21 years)	1,449 subjects aged 65-80 years	Evaluation of dietary intakes with a 208-item FFQ	The Mayo Clinic AD Research Center criteria were applied for diagnosing MCI; MMSE, CFT, PPBT, LDST, episodic memory with immediate word recall tests; executive function with the Stroop test, and prospective memory with a task by Einstein	Elevated SFA intake at midlife was associated with poorer global cognitive function and prospective memory and with an increased risk of MCI. High intake of PUFA was associated with better semantic memory. Also frequent fish consumption was associated with better global cognitive function and semantic memory. Further, higher PUFA/SFA ratio was associated with better psychomotor speed and executive function

(Table 1) contd....

Reference	Setting and Study Design	Subjects	Dietary Assessment	Cognitive Outcomes	Results and Conclusions
Dietary Fatty Acids, Fish Consumption and Dementia Syndromes					
Engelhart <i>et al.</i> , 2002 [61] The Rotterdam Study The Netherlands	Longitudinal, population-based (6 years)	5,395 subjects aged 55 and older	Evaluation of dietary intakes with a 100-item FFQ	Diagnosis of dementia (DSM-III-R criteria), AD (NINCDS-ADRDA criteria), and VaD (NINCDS-AIREN criteria)	High intakes of total fat, saturated fat, <i>trans</i> fat, and cholesterol and low intake of MUFA, PUFA, n-6 PUFA, and n-3 PUFA were not associated with an increased risk of dementia, AD, or VaD
Luchsinger <i>et al.</i> , 2002 [62] Washington Heights-Inwood Columbia Aging Project USA	Longitudinal, population-based (4 years)	980 subjects, mean age: 75.3±5.8 years	Evaluation of dietary intake with a 61-item FFQ	Diagnosis of prevalent dementia (DSM-IV criteria) and incident AD (NINCDS-ADRDA criteria)	Higher intake of calories and fats may be associated with higher risk of AD in subjects carrying the APOE ε4 allele
Morris <i>et al.</i> , 2003 [65] Chicago Health and Aging Project (CHAP) USA	Longitudinal, population-based (3.9 years)	815 subjects, aged 65 years and older	Evaluation of dietary intake with a 154-item FFQ	Incident diagnosis of AD (NINCDS-ADRDA criteria)	Higher intake of n-3 PUFA and weekly fish consumption may reduce the risk of incident AD
Huang <i>et al.</i> , 2005 [72] Cardiovascular Health Cognition Study (CHCS) USA	Longitudinal, population-based (5.4 years)	5,201 participants, aged 65 years and older	Evaluation of dietary intake with a 99-item FFQ	Incident diagnosis of dementia (DSM-IV criteria) and AD (NINCDS-ADRDA criteria)	Fatty fish, such as tuna or “other fish” was associated with a lower risk of developing dementia and AD with a dose-response relationship, whereas lean, fried fish was not. Those without an APOE ε4 allele had a 35 to 45% lower risk with consumption of fatty fish, whereas there was little or no difference for APOE ε4 allele carriers
Laitinen <i>et al.</i> , 2006 [66] Cardiovascular Risk Factors Aging and Dementia (CAIDE) Finland	Longitudinal, population-based (21 years)	1,449 subjects aged 65-80 years	Evaluation of dietary intake with a 154-item FFQ	Incident diagnosis of AD (NINCDS-ADRDA criteria)	Moderate intake of PUFA at midlife was protective, whereas a moderate intake of SFA may increase the risk of dementia and AD, especially among APOE ε4 carriers.
Barberger-Gateau <i>et al.</i> , 2007 [67] Three-City Study France	Longitudinal, population-based (4 years)	9,294 subjects, aged 65 years and older	Evaluation of dietary intakes with a FFQ	Incident diagnosis of dementia (DSM-IV criteria) and AD (NINCDS-ADRDA criteria)	Frequent consumption of fruits and vegetables, fish, and n-3 rich oils may decrease the risk of dementia and AD, especially among APOE ε4 non-carriers

MMSE = Mini-Mental State Examination (global cognitive functioning); PUFA = polyunsaturated fatty acids; FFQ = food frequency questionnaire; PMSQ = Pfeiffer's Mental State Questionnaire (global cognitive functioning); EBMT = East Boston Memory test (immediate and delayed episodic memory); SDMT = Symbol Digit Modalities Test (perceptual-motor speed); MUFA = monounsaturated fatty acids; CCI = Charlson comorbidity index; BMI = body mass index; SMC = subjective memory complaint; ADL = activities of daily living; IADL = instrumental activities of daily living; CFT = Category Fluency Test (semantic memory); PPBT = Purdue Peg Board task (psychomotor speed) LDST = Letter Digit Substitution Test (perceptual-motor speed); DSM-III-R: Diagnostic and Statistical Manual of Mental Disorders, third edition revised; NINCDS-ADRDA: National Institute of Neurological and Communicative Disorders and Stroke and the Alzheimer's Disease and Related Disorders Association; NINCDS-AIREN: National Institute of Neurological and Communicative Disorders and Stroke and the Association Internationale pour la Recherche et l'Enseignement en Neurosciences; DSM-IV: Diagnostic and Statistical Manual of Mental Disorders, fourth edition; APOE = apolipoprotein E; SFA = saturated fatty acids.

have found no association [49]. A lower risk of incident dementia in those subjects consuming more fish has nevertheless been reported in several other independent prospective cohort studies [63, 65, 67, 72]. Therefore, despite these contrasting findings, there is now considerable evidence suggesting that fatty acid intake and fish consumption may influence dementia and AD risk, but the direction (protection or risk) and the level of this effect remain unclear [33]. Finally, some recent randomized controlled trials (RCTs) assessed the cognitive or functional effect of n-3 PUFA sup-

plementation on patients with VaD, AD, MCI, or ARCD in cognitively unimpaired older subjects [73-79]. These RCTs suggested a positive effect of this intervention only in very mild AD or MCI patients, or in subgroups [e.g., APOE-ε4 carriers] for cognitive performance in nondemented subjects or for neuropsychiatric symptoms in mild to moderate AD patients [73]. Supplemental DHA may not necessarily be recommended when the decline is more severe, possibly because additional DHA might contribute to degenerative processes in the brain related to lipid peroxidation [80].

In conclusion, epidemiological evidence suggested a possible association among fish consumption, MUFA, and PUFA (particularly, n-3 PUFA) and reduced risk of cognitive decline and dementia. However, due to the small number of studies that inform this topic, further research is necessary before a strong conclusion can be drawn. Based on the current evidences from human and animal studies, Friedland concluded that, currently, it is not possible to make definitive dietary recommendations in relation to the AD risk on fish consumption and the lower intake of saturated fat from meat and dairy products [81] as well as on UFA consumption or lower intake of saturated fat in relation to the risk for dementia and cognitive decline. However, a high consumption of fats from fish, vegetable oils, vegetables, and nuts should be encouraged because this dietary advice is in accordance with recommendations for lowering the risk of cardiovascular disease, obesity, diabetes, and hypertension [33, 81].

Alcohol Consumption in Predementia and Dementia Syndromes

Recent studies with older samples have indicated that a U- or J-shaped curve may best describe the relation between level of alcohol consumption and cognitive performance [82-86]. As a general statement, drinkers of light to moderate amounts of alcohol performed at a higher cognitive level than abstainers or heavy drinkers [37], with some notable exceptions [83, 86]. However, definitions describing how much alcohol is in a drink vary remarkably by article and country, and terms such as light, moderate, and heavy drinking are variably defined [37]. Lower levels of alcohol intake have proportionally greater effects in the elderly, due to their decreased lean body mass and lower percentage of body weight made up of water and a reduced activity of alcohol dehydrogenase (ADH) and aldehyde dehydrogenase (ALDH). In fact, in several cross-sectional studies, moderate drinking, from up to 1 drink per day (up to 14 g of alcohol) to 4 drinks per day (52 g of alcohol), as compared with non-drinking has been associated with a better function in many cognitive domains [87-90].

Furthermore, in cross-sectional studies, both better [91-95] and poorer [96] cognitive performances with higher levels of alcohol drinking have been observed. The studies that have assessed changes in cognition prospectively have had contradictory results [37] (Table 2). Drinkers have been proposed to have a greater decrease in global cognitive function [97] and attention [98] as compared with nondrinkers, but moderate drinkers (about one drink or less than 15 ml of alcohol per day) to have less decline in general cognition [99] or psychomotor speed [100] than nondrinkers. Furthermore, some studies showed no association at all [93, 101-103]. In particular, the associations between current cognitive performance and alcohol drinking 5-24 years earlier have also been studied, with contrasting results. In fact, the global cognitive function may be poorer among both alcoholics and past drinkers as compared with nondrinkers or moderate drinkers [84] and better among moderate drinkers and poorer among heavy drinkers as compared with nondrinkers [104]. Moreover, smoking or APOE $\epsilon 4$ allele often modified the association between alcohol drinking and cognitive functions. However, studies on the interaction between smoking and alcohol drinking have yielded inconclusive

results [92, 98, 105]. The role of alcohol for the cognition might be modified by the presence of the APOE $\epsilon 4$ allele [88, 97], but the patterns of interaction suggested have been different results [99, 105]. Some studies investigating the effect of different types of alcohol on the cognitive function found no beverage-specific differences [84, 99, 105].

Alcohol drinking is also one of the possible determinants of clinical dementia [37, 98, 100, 102, 106]. In fact, epidemiological studies have reported an association between red wine consumption and the incidence of AD and dementia in cross-sectional and longitudinal studies [107, 108] (Table 2). These results were consistent with the findings from other cohort studies [109-112], but not with others [98, 100, 102, 106, 113, 114]. Some other population-based, prospective studies, with longer follow-up periods studied the effects of different patterns of alcohol intake on dementia [115-117]. For predementia syndromes, in 1,445 non-cognitively impaired individuals aged 65 to 84 years from the ILSA, patients with MCI who consumed up to 1 drink per day of alcohol had a reduction in the rate of progression to dementia in comparison with patients with MCI who never consumed alcohol. Overall, vs non-drinkers, patients with MCI who consumed less than 1 drink per day of wine had a decrease in the rate of progression to dementia of about 85%. No significant associations were found between any levels of drinking and the incidence of MCI in non-cognitively impaired individuals vs. abstainers [118]. Only a few other studies have examined the effect of alcohol consumption on the risk for the incidence of MCI [119, 120]. After an average follow-up of 23 years, non-drinkers and frequent drinkers were both more than twice as likely to have MCI in old age than occasional drinkers [119]. However, the APOE genotype seemed to modify the relationship, such that the risk of old age dementia increased with increasing midlife alcohol consumption only among carriers of the APOE $\epsilon 4$ allele [119]. On the other hand, the findings from the ILSA were consistent with those obtained in the Women's Health Initiative Memory Study with a 4.2 year-follow-up, which found that moderate alcohol intake (≥ 1 drink per day) was associated with an approximately 50 percent reduced risk of combined probable dementia and MCI [120]. However, after adjusting for demographic and socioeconomic factors, and baseline Modified Mini Mental State Examination (3MSE), the significance disappeared [120].

Very recently, a systematic review with meta-analysis was carried out to investigate any relationship between incident cognitive decline or dementia in the elderly and alcohol consumption between 1995 and March 2006, with only longitudinal studies of subjects aged ≥ 65 included [121]. Definitions of outcomes varied with some studies reporting on dementia, or AD alone, or AD and dementia, or VaD, or AD and VaD, or all dementia, VaD and AD [121]. This meta-analysis suggested that, at least in epidemiological studies, light to moderate alcohol use was associated with a 38% reduced risk of unspecified incident dementia. Also in the studies included in this meta-analysis, there was no close agreement as to the optimal level of alcohol consumption and the classification of light to moderate drinking varied very widely. In fact, for dementia, benefit was shown for more than 1 drink per day, weekly or monthly wine consumption, 250-500 ml (usually wine) or more than are

Table 2. Principal Longitudinal Clinical and Epidemiological Studies on the Relationships between Alcohol Consumptions and Dementia (i.e., Alzheimer' Disease, AD, and Vascular Dementia, VaD) or Predementia Syndromes (i.e., Age-Related Cognitive Decline, ARCD, and Mild Cognitive Impairment, MCI)

Reference	Setting and Study design	Subjects	Methods	Results and Conclusions
Alcohol Consumption and Predementia Syndromes				
Launer <i>et al.</i> , 1996 [86] The Zutphen Elderly Study The Netherlands	Cross-sectional and longitudinal, population-based (3 years)	489 men aged 69-89 years	MMSE, evaluation of alcohol intake and smoking habits	Men with CVD/diabetes and low-to-moderate alcohol intake had a significantly lower risk for poor cognitive function than abstainers. Alcohol intake was not associated with cognitive decline
Elias <i>et al.</i> 1999 [87] The Framingham Heart Study USA	Cross-sectional and longitudinal, population-based (24 years)	1,786 subjects, aged 55-88 years	Eight cognitive tests of verbal memory, learning, visual organization and memory, attention, abstract reasoning, and concept formation; evaluation of weekly alcohol intake	Women who drank moderately (2-4 drinks/day) showed superior performance in many cognitive domains relative to abstainers. For men, superior performance was found within the range of 4-8 drinks/day, although fewer significant relations were observed. These results were confirmed by prospective analyses of 24-year drinking history
Dufouil <i>et al.</i> , 2000 [91] Epidemiology of Vascular Aging (EVA) Study France	Longitudinal, population-based (4 years)	1,389 subjects, aged 59-71 years	MMSE, evaluation of alcohol consumption, APOE genotyping, and smoking habits	Alcohol consumption was associated with a decreased risk of cognitive decline in individuals without the APOE $\epsilon 4$ allele, whereas moderate drinking increased the risk of decline in APOE $\epsilon 4$ allele carriers. Also, lifetime smoking was a risk factor for cognitive decline in individuals without the APOE $\epsilon 4$ allele. The data also suggested a slight protective effect of smoking in APOE $\epsilon 4$ allele carriers
Ngandu <i>et al.</i> , 2007 [105] Cardiovascular Risk Factors Aging and Dementia (CAIDE) Finland	Longitudinal, population-based (21 years)	1,341 participants aged 65-79 years	MMSE, and neuropsychological tests evaluating episodic memory, semantic memory, psychomotor speed, executive function, prospective memory, and subjective memory. Evaluation of alcohol intake, smoking habits, and APOE genotyping	The nondrinkers both at midlife and later had a poorer cognitive performance than drinkers, especially in the domains related to fluid intelligence, i.e., executive function, psychomotor speed, as well as episodic memory, whereas the other cognitive functions showed little association with alcohol drinking. No interactions between APOE $\epsilon 4$ and alcohol or sex and alcohol were found.
Anttila <i>et al.</i> , 2004 [119] Cardiovascular Risk Factors Aging and Dementia (CAIDE) Finland	Longitudinal, population-based study (23 years)	1464 men and women aged 65-79 years	Diagnosis of incident dementia and MCI, and subjects classified as those who never drank alcohol, those who drank "infrequently" (less than once a month), and those who drank "frequently" (several times a month)	Alcohol drinking in middle age showed a U shaped relation with risk of MCI in old age. Only the carriers of APOE $\epsilon 4$ had an increased risk of dementia with increasing alcohol consumption
Espeland <i>et al.</i> , 2005 [120] Women's Health Initiative Memory Study USA	Longitudinal, population-based study (4.2 years)	4,451 community-dwelling women aged 65-79 years	Diagnosis of incident MCI and dementia, and annual modified MMSE. Subjects classified as abstainers, < 1 drink/day, and > 1 drink/day	Moderate alcohol intake was associated with an approximately 50 percent reduced risk of combined probable dementia and/or MCI. Significant decline in global cognition was less common among women who reported alcohol intake at baseline, also after adjustment for both demographic/social and clinical factors
Solfrizzi <i>et al.</i> , 2007 [118] The Italian Longitudinal Study on Aging (ILSA) Italy	Longitudinal, population-based study (3.5 years)	1,445 men and women aged 65-84 years	Diagnosis of incident MCI and progression from MCI to dementia; subjects classified as abstainers, 0-1 drink/day, 1-2 drink/day, and > 2 drink/day	In patients with MCI up to 1 drink/day of alcohol or wine may decrease the rate of progression to dementia. No significant associations were found between any levels of drinking and the incidence of MCI in non-cognitively impaired individuals vs abstainers

(Table 2) contd.....

Reference	Setting and Study design	Subjects	Methods	Results and Conclusions
Alcohol Consumption and Dementia Syndromes				
Yoshitake <i>et al.</i> , 1995 [106] The Hisayama Study Japan	Longitudinal, population-based (7 years)	828 subjects aged 65 and over	Diagnosis of dementia, AD, and VaD; evaluation of alcohol consumption	Multivariate analysis showed that age, systolic blood pressure, drinking habits, and prior stroke episodes were significant independent precursors of VaD
Orgogozo <i>et al.</i> 1997 [107] Personnes Agees QUID (PAQUID) Study France	Longitudinal, population-based (3 years)	3777 subjects aged 65 and over	Diagnosis of dementia and AD; evaluation of alcohol intake	In the 318 moderate drinkers, as compared to the 971 non-drinkers, and after adjusting for possible confounders, the OR were respectively 0.19 for incident dementia and 0.28 for AD. Among the 922 mild drinkers respect to the abstainers the relative risk for AD was reduced significantly
Ruitenberg <i>et al.</i> , 2002 [109] The Rotterdam Study The Netherlands	Longitudinal, population-based study (6 years)	5,395 subjects aged 55 years and older	Diagnosis of AD, VaD, or other dementia; evaluation of alcohol intake, and APOE genotyping	Light-to-moderate drinking (1 to 3 drinks per day) was significantly associated with a lower risk of any dementia and VaD. No evidence that the relation between alcohol and dementia varied by type of alcoholic beverage was found
Mukamal <i>et al.</i> , 2003 [237] The Cardiovascular Health Study (CHS) USA	Nested case-control of a longitudinal, population-based study (6 years)	373 cases with incident dementia and 373 controls	Diagnosis of incident dementia (AD and VaD), average alcohol consumption, and magnetic resonance imaging findings	Compared with abstention, consumption of 1 to 6 drinks weekly is associated with a lower risk of incident dementia among older adults. A trend toward greater odds of dementia associated with heavier alcohol consumption was most apparent among men and participants with an APOE $\epsilon 4$ allele, with similar relationships of alcohol use with AD and VaD
Deng <i>et al.</i> , 2006 [112] Chongqing City Study China	Longitudinal, population-based study (2 years)	3,286 subjects, aged 60 years and older	Diagnosis of incident dementia, AD, and VaD and evaluation of alcohol intake	Light to moderate drinkers had a lower risk of dementia than nondrinkers, but a nonsignificant increased risk was observed in heavy drinkers
Mehlig <i>et al.</i> , 2008 [117] The Prospective Population Study of Women in Göteborg Sweden	Longitudinal, population-based study (34 years)	1,462 women aged 38–60 years	Diagnosis of dementia; evaluation of alcohol intake and smoking habits	Wine was protective for dementia, and the association was strongest among women who consumed wine only. In contrast, consumption of spirits at baseline was associated with slightly increased risk of dementia

MMSE: Mini-mental State Examination; CVD: cerebrovascular disease; APOE: apolipoprotein E; OR: odds ratio..

equivalent to 3 drinks per day and from 1–28 units per week [37]. For AD, light to moderate alcohol was associated with a significantly reduced risk of 32%, defining optimal amounts as weekly consumption of wine, 1–6 units or more than 2 drinks per week, or more than 3 drinks/250–500 ml per day (usually wine), or where studied by gender, 1–3 per day in males [37]. Although the point estimates were also in a similar direction for VaD and cognitive decline (0.82 and 0.89 respectively), the results were not statistically significant [121]. With regard to cognitive function, results for optimal consumption were mixed, either above/below or equal to 1 drink a month or day, in subjects with cardiovascular disease or diabetes, 1–2 drinks per week, while for VaD, 1–3 drinks per day in males appeared to be beneficial [37]. Finally, another very recent meta-analysis included 15 prospective studies (follow-ups ranged from 2 to 8 years), with samples including 14,646 participants evaluated for AD, 10,225 participants evaluated for VaD, and 11,875 followed

for any type of dementia [122]. This meta-analysis indicated that light to moderate alcohol intake was associated with a 25%–28% reduction in risk of AD, VaD, and any dementia compared with alcohol abstinence in older adults. Heavy drinking was not associated with increased risk of dementia in the present study [122].

In conclusion, low to moderate alcohol drinking has been proposed as a protective factor against MCI and dementia in several longitudinal studies, but contrasting findings also exist [37]. In fact, many of these studies were limited by cross-sectional design, restriction by age or sex, or incomplete ascertainment. Moreover, different outcomes, beverages, drinking patterns, or follow-up periods, or possible interactions with other lifestyle-related (i.e., smoking) or genetic factors (i.e., APOE genotyping) may be sources of great variability. In fact, the body of evidence within the last 10 years suggested that low to moderate alcohol use may be associated with a reduced risk of unspecified incident de-

mentia and AD, while for VaD, cognitive decline, and pre-dementia syndromes the current evidence is only suggestive of a protective effect [37, 115]. Therefore, at present, the preferable outcomes for this kind of studies appear to be unspecified dementia and AD, given the small number of studies reporting VaD as an outcome and the difficulties in the classification of cognitive decline and pure VaD, often including cases with vascular factors in the AD category. On the other hand, the cardiovascular mechanisms behind the protective effects of alcohol may have an even greater effect on VaD.

For the different types of beverages, several studies suggested that low to moderate wine consumption may be protective against dementia and cognitive decline and not total alcohol intake, beer, or spirits, with strong but not conclusive evidence [37]. In this context, it would be desirable to differentiate red and white wine, but this information is not currently available. Whether the present divergent findings can be explained by the drinking patterns has not been extensively investigated, and studies with long follow-up periods are few [37]. Therefore, the role of alcohol drinking for cognitive functions might be somewhat different at different time points. Nonetheless, the current evidence suggested that alcohol consumption in old age was substantially in agreement with midlife consumption in relation to cognitive decline [37]. At present, there is no evidence indicating that starting to drink at a later age would be beneficial against predementia or dementia syndromes. In conclusion, as intervention studies are not feasible in this area, the best evidence comes from an overview of epidemiological studies. At present, there is no indication that low to moderate alcohol drinking would be harmful to cognition and dementia. However, it is not possible to define a specific beneficial level of alcohol intake in relation to cognitive functions and dementia. Taking into account the well-known harms related to heavy alcohol drinking and the lack of long-term follow-up studies or randomized controlled trials, it would be too early to recommend alcohol to abstainers to preventing cognitive decline or predementia syndromes or delaying the onset of dementia. Probably, at present, in moderate to heavy alcohol drinkers, experiencing memory difficulty, or with suggested diagnoses of MCI or early AD [123], among management options, given the challenge in achieving total abstinence, a viable alternative option could also be low to moderate drinking [124].

High Intakes of Cereals, Fruits, and Vegetables in Cognitive Decline

Some findings showed that older African-Americans [125] and Japanese living in the United States [126] have much higher prevalence of AD (6.24% and 4.1%, respectively) than those still living in their ethnic homelands (<2%) suggested that the prevalence of AD is more strongly influenced by diet and nutrition, environment, and/or lifestyle, than by genetics. However, the simple comparison of prevalence data might be misleading, since there is a large difference in selective survival and other cultural effects [23]. A more valuable comparison would be that of the overall dementia rate between different countries, culturally adjusted. In a cross-sectional study on AD, regression analyses have been performed in the 65+ age population of 11 countries ob-

tained from 18 community-wide studies versus components of the national diets [23, 24]. The primary finding is that fat and total caloric supply have the highest correlations with AD prevalence rates. A significant inverse correlation was found between the fraction of calories derived from cereals and AD prevalence [23-26, 127, 128]. While whole grains have antioxidant vitamins and minerals, it is not clear whether the cereals generally consumed are whole grains. Therefore, this relationship could be due to the fact that countries with low fat supply have high cereals supply, rather than to a direct therapeutic effect of the cereals. These ecological findings on the possible role of cereals as a risk reduction factor were confirmed in a 4-year cohort study in New York, the Washington Heights-Inwood Columbia Aging Project (WHICAP), that also found that dietary fat was an important risk factor for AD for those with the APOE $\epsilon 4$ allele, but not for those without that allele [129].

Given the possible role of vascular factors in contributing to cognitive decline [11, 13], high intake of fruits and vegetables appeared to decrease the risk of CAD [130] and stroke [131], and thus may decrease also cognitive decline. In fact, bioactive compounds like antioxidants, mostly contained in fruits and vegetables, are important for the protection against oxidative and nitrosative stress and these micronutrients have been related with aging itself and the pathophysiology of some age-related illnesses including cognitive impairment and dementia [132]. While a growing body of knowledge demonstrates both the importance of oxidative stress in the aetiology of dementia and the efficacy of antioxidant treatment in animal and cellular models, studies in humans are presently inconclusive [133, 134]. In fact, epidemiological studies have shown that dietary intake of natural or synthetic products with a putative antioxidant effect, mainly vitamin E, improve cognitive function and reduce the risk of AD [135, 136]. Moreover, antioxidative intervention studies in animal models of AD-like amyloidosis show a significant reduction in oxidative stress, A β deposition and behavioral improvements [137, 138]. However, intervention trials of antioxidant supplementation have demonstrated no major benefit against cognitive impairment [134, 139]. In AD clinical trials, antioxidants have shown only a marginal positive effect on disease progression [140, 141], and a recent MCI trial with antioxidants, indicates that vitamin E ingestion for three years has no benefit on the risk of progression to AD [142]. There are several reasons explaining this discrepancy. As recently reviewed [134, 143], the failure of these trials likely arises from a combination of factors, including not monitoring the drug levels and surrogate markers for the *in vivo* therapeutic effect of the drug of interest and starting the therapy very late in the disease stage. In accordance with convergent data, oxidative stress is a very early event in AD, occurring in pre-symptomatic phases, and so antioxidant strategies should start years before the dementia age risk.

The limited epidemiological evidence available on fruit and vegetable consumption and cognition generally supported a protective role of these macronutrients against cognitive decline (Table 3). Associations between greater intake of fruits and vegetables and better cognitive performance have been found in two cross sectional studies [34, 144]. In the Nurses' Health Study, when fruit and vegetable intake in

Table 3. Principal Longitudinal Clinical and Epidemiological Studies on the Relationships between, Fruits and Vegetables Intake and Dementia (i.e., Alzheimer' Disease, AD) or Predementia Syndromes (i.e., age-Related Cognitive Decline, ARCD)

Reference	Setting and Study Design	Subjects	Methods	Results and Conclusions
Kang <i>et al.</i> , 2005 [145] The Nurses' Health Study USA	Longitudinal, population-based (10-16 years)	13,388 women aged 70 years and over	The TICS and other five cognitive tests (immediate and delayed recalls of the EBMT, category fluency, delayed recall of the TICS, and DSB) and evaluation of dietary intake of food with a semi-quantitative FFQ	High consumption of vegetable, but not fruit, was associated with less cognitive decline among older woman
Morris <i>et al.</i> , 2006 [146] Chicago Health and Aging Project (CHAP) USA	Longitudinal, population-based (3-6 years)	3,718 subjects, aged 65 years and older	Immediate and delayed recalls of the EBMT, MMSE, and SDMT and evaluation of dietary intake of food with a semi-quantitative FFQ	High vegetable but not fruit consumption was associated with slower rate of cognitive decline with older age
Dai <i>et al.</i> , 2006 [34] The Kame Project USA	Longitudinal, population-based (6.3 years)	3045 Japanese Americans subjects, aged 65 or over	Diagnosis of incident AD and evaluation of dietary intake of food with a semi-quantitative FFQ	Frequent drinking of fruit and vegetable juices was associated with a substantially decreased risk of AD. This inverse association was stronger after adjustments for potential confounding factors
Hughes <i>et al.</i> , 2009 [148] The HARMONY study Sweden	Longitudinal (30 years)	3,779 members of the Swedish Twin Registry	Diagnosis of incident dementia and AD and evaluation of dietary habits	These findings suggested that higher fruit and vegetable intake assessed at midlife was associated with a lower risk of dementia and AD later

TICS = Telephone Interview for Cognitive Status; EBMT = East Boston Memory Test; DSB = Digit Span Backward; FFQ = food frequency questionnaire; MMSE = Mini-Mental State Examination; SDMT = Symbol Digit Modalities Test

relation to cognitive function and decline among aging women were prospectively followed from 1976 to 2003 with biennial telephone questionnaires, women in the highest quintile of cruciferous vegetables declined slower compared with the lowest quintile [145]. In the same study, women consuming the highest quantity of green leafy vegetables also experienced slower decline than women consuming the least amount [145]. These findings were consistent with those from a subset of subjects participating to the Chicago Health and Aging Project (CHAP) [146], a high vegetable consumption was associated with a slower rate of cognitive decline over 6 years. Of the different types of vegetables, green leafy vegetables had the strongest association. Fruits consumption was not associated with cognitive change [146]. High intake of beta carotene, flavonoids, vitamins C and E, thiamine, and folate from dietary fruit and vegetables was also associated with a lower risk of AD in the Rotterdam Study [147]. Furthermore, in the Kame Project cohort, a population-based prospective study on 1,836 Japanese Americans aged 65 or older in King County, Washington, after an average of six years, follow-up surveys found that 63 participants had developed AD. After adjusting for age, sex, education and other relevant factors, those who reported drinking at least three glasses of juice a week had a 76 percent lower risk of developing AD than did those who reported drinking juice less than once a week [34]. Data from the 3C Study demonstrated that a diet rich in fish, n-3 PUFA rich oils, fruits, and vegetables could contribute to decrease the risk of dementia and AD in older persons whereas consumption of n-6 rich oils could exert detrimental effects

when not counterbalanced by sufficient n-3 intake. These effects seem more pronounced among APOE $\epsilon 4$ noncarriers [67]. Finally, among the 3,779 members of the Swedish Twin Registry who completed a diet questionnaire approximately 30 years before cognitive screening and full clinical evaluation for dementia as part of the study of dementia in Swedish Twins (HARMONY) study, a medium or great proportion of fruits and vegetables in the diet, compared with no or small, was associated with a decreased risk of dementia and AD. This effect was observed among women and those with angina [148].

One important unanswered question relates to the largely unexplored relationship among intake of fruits and vegetables, antioxidant micronutrient status, a condition of oxidative stress, and cognitive performance in healthy subjects [149]. In a very recent cross-sectional study, 193 healthy community dwellers consuming a diet rich in fruits and vegetables had higher plasma levels of lipophilic antioxidant micronutrients, lower levels of biomarkers related to oxidative stress, and better scores on neuropsychological evaluation compared to subjects with low intakes of fruits and vegetables. This result was independent of gender, education, BMI, lipid profile, and albumin levels but, most importantly, was independent of age. Therefore, micronutrient levels appeared to be a good indicator for fruit and vegetable intake and important in influencing the *in vivo* oxidant/ antioxidant balance in healthy subjects [150]. Furthermore, individuals who consume a diet high in fruits and vegetables as well as nuts or fish would probably have other lifestyle char-

acteristics that might be effective in reducing AD or VaD risk (e.g., physical activity) [151].

THE WHOLE-DIET APPROACH TO THE MEDITERRANEAN DIET MODEL

In the last years, there is growing interest in exploring whole types of foods (eg, tomatoes, fruits, and vegetables) and dietary patterns. In fact, MeDi could be an interesting model to further study the association between dietary patterns and cognitive functioning, given the suggested role of many components of this diet (MUFA, n-3 PUFA, fish and olive oil consumption, cereals, and red wine) in contrasting cognitive impairment. The evidence reviewed thus far is based on the hypothesis that individual nutrients contained in foods may have effects on predementia and dementia syndromes, neglecting the possible importance of the additive or synergistic effects of other nutrients within and across particular foods that form part of a diet [33].

However, as seen extensively above, the results about isolated nutrient or food appear to be conflicting. Compared with traditional single-food or nutrient methods, the whole-diet or dietary pattern approach is appealing for several reasons. Indeed, the analyses of single nutrients ignore important interactions (additive, synergistic or antagonist effects) between components of diet and more importantly people did not eat isolated nutrients [152, 153]. The evaluation of the adherence to a Mediterranean-style diet was conducted to develop scores or indexes called 'Mediterranean Diet Scale' or 'Mediterranean Diet Score' or 'Mediterranean Adequacy Index' or even 'Mediterranean-Style Dietary Pattern Score' based on a priori hypotheses and intended to be close to the traditional one [152, 154, 155]. In particular, a scale indicating the degree of adherence to the traditional MeDi was constructed by Trichopoulou and colleagues [156], revised to include fish intake [157], and used in recent studies about cognitive decline on US, French, and Greek cohorts [56, 158-163]. A value of 0 or 1 was assigned to each of nine indicated components (vegetables, fruits, legumes, cereals, the ratio of MUFA to SFA, alcohol and fish presumed to be beneficial; meat and dairy products presumed to be detrimental) with the use of the sex-specific median as the cut-off. The total MeDi score ranged from 0 (minimal adherence to the traditional MeDi) to 9 (maximal adherence).

The use of diet-scoring systems such as the MeDi score have undeniable advantages in understanding the role of diet in chronic disease [164]. Scarmeas and colleagues, using data from the WHICAP with an inclusive dietary score (MeDi score) but studying a population with a substantial difference in dietary habits in comparison with Greek (European Prospective Investigation into Cancer and Nutrition, EPIC) [56], Italian (ILSA) [54, 55], and French (3C Study) [162] cohorts, found that higher adherence to the MeDi is associated with a trend for reduced risk of incident MCI and also with reduced risk of MCI progression to AD [161] (Table 4). Individuals older than 65 years of age were enrolled in the WHICAP in 1992 and 1999 and have been followed on average for 4 years. Among individuals free from dementia at baseline, 2258 individuals had a complete assessment of cognitive functions and of dietary habits (Table 2). These findings confirmed previous reports from the WHICAP in

which higher adherence to the MeDi reduced the risk for getting AD [158, 159], and was associated with lower mortality in AD [160] (Table 4). They may account for the complex biologic interactions between different components of a composite dietary pattern, such as MeDi, that may be difficult to detect in analyses focusing on the only individual components [165]. The potentially beneficial effect of the MeDi for cognitive decline may be the result of some of its individual food components. For example, as seen above, potentially beneficial effects for MCI, its progression to dementia, and AD or dementia have been reported for alcohol [37], fish consumption [33], PUFA [33], and lower SFA [33]. Interestingly, in other studies, alcohol [37] and PUFA [33] have failed to be associated with protection against MCI, its progression to dementia, and AD or dementia. These contrasting findings about the impact of MeDi score or individual macronutrients on predementia and dementia syndromes may suggest an approach not confined only to cognitive skills but extended to functional status and comorbidity. However, we should not renounce a priori to work for a correct estimate of the validity of the MeDi score for cognitive impairment as a criterion. However, while several prospective cohort studies have consistently indicated that adherence to the traditional MeDi is associated with longevity or specific health outcomes, no study has investigated the relative importance of individual components of the MeDi score in the generation of this association. Very recent findings from the EPIC on mortality suggested that the contribution of the nine components of the MeDi to the apparent protective effect of the score assessing adherence to traditional Mediterranean diet is approximately additive. The dominant components of the MeDi score as a predictor of lower mortality were moderate consumption of alcohol, low consumption of meat and meat products, and high consumption of vegetables, fruits and nuts, olive oil, and legumes [166]. However, the evidence about the role of the whole MeDi on cognitive decline are still scarce [152]. Therefore, in future studies, it would be indicated, along with measuring this effect by a dietary composite score, also to report the estimates and the impact of the individual components of the diet (particularly fats) vs. whole-diet approach, to better understand if the whole MeDi is likely important and not only sources of beneficial fats (e.g., MUFA and n-3 PUFA). At present, only in the EPIC cohort has been studied a whole-diet approach plus individual nutrients analysis with no significant association between MeDi score and global cognitive scores, whereas a statistically significant inverse association was found between cognitive performance and some individual dietary components [56]. In fact, in a total of 732 men and women, 60 years or older, participating in the EPIC-Greece cohort and residing in the Attica region, six to 13 years later, seed oil consumption or PUFA intakes may adversely affect cognition, whereas adherence to the MeDi, as well as intake of olive oil, MUFA and SFA exhibited weakly positive but not significant associations [56] (Table 4). In a very recent re-analysis from the ILSA cohort, we showed that high PUFA were associated with reduced risk of incident MCI among who consumed a low MUFA/SFA ratio intake [167]. In fact, while an increasing body of evidence suggested that elevated fish consumption and high intake of n-3 PUFA may be protective against ARCD and MCI, the traditional Cretan diet, although strongly dependent on high

Table 4. Principal Epidemiological Studies on the Relationship Among Adherence to the Mediterranean Diet (MeDi) and Dementia (i.e., Alzheimer' Disease, AD) or Predementia Syndromes (i.e., Age-Related Cognitive Decline, ARCD, and Mild Cognitive Impairment, MCI) In older People.

Reference	Setting and Study Design	Subjects	Dietary Assessment	Cognitive Outcomes	Results and Conclusions
Scarmeas <i>et al.</i> , 2006 [158] Washington Heights-Inwood Columbia Aging Project (WHICAP) USA	Longitudinal, population-based (4 years)	2,258 subjects, aged 65 and over	Evaluation of dietary intake with a 61-item FFQ. A dietary composite score (MeDi score) evaluated adherence to MeDi	Diagnosis of incident AD	Higher adherence to MeDi reduced risk for probable AD either with or without coexisting stroke
Scarmeas <i>et al.</i> , 2007 [160] Washington Heights-Inwood Columbia Aging Project (WHICAP) USA	Longitudinal, population-based (4.4 years)	192 community-based individuals aged 65 and older who were diagnosed with AD	Evaluation of dietary intake with a 61-item FFQ. A dietary composite score (MeDi score) evaluated adherence to MeDi	All-cause mortality	Higher adherence to the MeDi was associated with lower mortality in AD. The gradual reduction in mortality risk for higher MeDi adherence tertiles suggested a possible dose-response effect
Psaltopoulou <i>et al.</i> , 2008 [56] European Prospective Investigation into Cancer and Nutrition (EPIC) Greece	Longitudinal, population-based (median 8 years)	732 subjects, aged 60 years or older	Evaluation of dietary intakes with a 150-item FFQ. A dietary composite score (MeDi score) evaluated adherence to MeDi	MMSE	No significant association between MeDi score and MMSE scores, whereas a statistically significant inverse association was found between MMSE performance and some individual dietary components, such as seed oil or PUFA intakes
Scarmeas <i>et al.</i> , 2009 [161] Washington Heights-Inwood Columbia Aging Project (WHICAP) USA	Longitudinal, population-based (4.5 years)	2,258 subjects, aged 65 and over	Evaluation of dietary intake with a 61-item FFQ. A dietary composite score (MeDi score) evaluated adherence to MeDi	Incidence of MCI and progression from MCI to AD	Higher adherence to the MeDi was associated with a trend for reduced risk of developing MCI and with reduced risk of MCI progression to AD
Scarmeas <i>et al.</i> , 2009 [163] Washington Heights-Inwood Columbia Aging Project (WHICAP) USA	Longitudinal, population-based (5.4 years)	2,247 subjects aged 65 and over	Evaluation of dietary intake with a 61-item FFQ. A dietary composite score (MeDi score) evaluated adherence to MeDi. Two slightly different versions of the Godin leisure time exercise questionnaire were also used	Diagnosis of incident AD	Both higher Mediterranean-type diet adherence and higher physical activity were independently associated with reduced risk for AD
Feart <i>et al.</i> , 2009 [162] Three-City Study France	Longitudinal, population-based (5 years)	1,410 adults subjects, aged 65 years and older	Evaluation of dietary intakes with a FFQ. A dietary composite score (MeDi score) evaluated adherence to MeDi	Cognitive performance was assessed with MMSE, IST, BVRT, and FCSRT. Incident cases of dementia were also diagnosed	Adherence to a Mediterranean type diet was associated with slower decline on the MMSE but not other cognitive tests and was not associated with the risk for incident dementia over 5 years of follow-up

FFQ = food frequency questionnaire; MMSE = Mini-Mental State Examination (global cognitive functioning); IST = Isaacs Set Test, BVRT = Benton Visual Retention Test; FCSRT = Free and Cued Selective Reminding Test

olive oil intake, was never centred on fish consumption [168]. In this context, n-6 PUFA could potentially exert some health benefits. In fact, in MeDi, some foods are rich in n-6 PUFA (e.g., walnuts, almonds, hazelnuts), while other foods, although poor in n-6 PUFA, are highly consumed

such as cereals, legumes, and, in less amount, some types of meat (pork), and poultry. Furthermore, olive oil contains n-6 and n-3 PUFA in a ratio of 10/1. Therefore, it should be advisable to include PUFA in the MeDi score as other individual macronutrient (i.e., MUFA/SFA ratio), among the com-

ponents presumed to be beneficial, in evaluating the relationship between adherence to MeDi and ARCD or MCI.

Very recently, a report from the 3C Study examined whether adherence to a MeDi was associated with change in cognitive performance and with lower risk of all-cause dementia or AD over a 5-year period in a sample of older community dwellers in France. Feart and colleagues concluded their report suggesting that higher adherence to a Mediterranean diet was associated with slower cognitive decline (MMSE score) while it was not associated with lower risk of incident dementia or AD [162] (Table 4). These findings were compared with those of the WHICAP from US on the possible role of adherence of a Mediterranean dietary pattern on the risk of MCI and AD [158, 161]. However, some concerns could be expressed about the analysis and discussion of the study findings. Our first concern was on the comparison between the French and US findings. In fact, from a methodological point of view, French data were left-truncated (in contrast with the US data), and the results of MeDi score of the 3C Study and WHICAP were non comparable (the medians used as cut-offs are generally study-specific). Moreover, from a clinical point of view, French findings were on ARCD which was measured by MMSE performance, US findings had incident MCI as primary outcome [161], which is a more structured clinical construct and not only a cognitive test. The second concern regards the role of drop-out dependence on outcome of interest that was not addressed, particularly because the slower MMSE performance across time was only borderline significant, and follow-up was over 5 years. Finally, Feart and colleagues correctly acknowledged among potential limitations of the study that, as already reported in the non-Mediterranean population of the WHICAP [158, 161], they found a relatively low ratio of MUFA to SFA. The hallmark of the traditional MeDi is a high consumption of olive oil, leading to a high ratio, in this study was verified that consumption of olive oil was positively correlated with the diet score and with the ratio (data available on request). Nonetheless, these positive associations did not give any indication about the adherence to the MeDi pattern as it can be made from the ratio of MUFA to SFA intake in the context of other parameters of the MeDi score. Finally, another original analysis from the WHICAP was the investigation of the risk of incident AD according to the adherence to the MeDi and to the practice of physical exercise, already known as a potential protective factor against functional decline [13]. Briefly, the authors observed that both high Mediterranean-type diet adherence and participating in physical activity were independently associated with a lower risk of AD over time [163] (Table 2).

MEDITERRANEAN DIET AND COGNITIVE DECLINE: POSSIBLE MECHANISMS

At present, no clear synergistic mechanism of the whole MeDi on predementia and dementia syndromes has been proposed, but different biological mechanisms could be evoked to explain the association between foods or nutrients of the MeDi and potential neuroprotective properties. First of all, vascular comorbidities, including diabetes, hypertension, dyslipidemia, and white matter lesions (WMLs), have been related to dementia, AD, and MCI [11, 14]. Furthermore, the role of nutrition on cardiovascular events was well docu-

mented [169], and there was strong evidence relating the MeDi to a lower risk of vascular risk factors such as dyslipidemia [170, 171], hypertension [170-172], and CAD [171, 173]. Therefore, a protective effect of the MeDi against VaD, the vascular component of AD, or MCI variants particularly related to cerebrovascular disease (CVD) (i.e., naMCI) [11, 174] could be expected.

Nevertheless, nonvascular biological mediating mechanisms (ie, metabolic, oxidative, and inflammatory) may be also responsible for the epidemiological MeDi-MCI associations [159]. In fact, although vascular variables are likely to be in the causal pathway between MeDi and AD and should be considered as possible mediators, when in the WHICAP vascular risk factors were considered in the models, the association between MeDi and AD did not change [159]. Furthermore, dementia and predementia syndromes and the MeDi have been associated with metabolic abnormalities. Insulin resistance and subsequent hyperinsulinaemia have been found to increase the risk of AD and promote decline in memory and cognitive dysfunction [175, 176]. Furthermore, MCI has been linked to dysregulation of glucose and insulin homeostasis [173, 174] and diabetes [175, 176]. At the same time, higher adherence to the MeDi seems to improve carbohydrate metabolism, and in interventional studies it has been associated with significant reductions in plasma glucose levels [171, 177], serum insulin levels, and insulin resistance [177, 178].

Oxidative stress could be another biological mechanism relating dementia, predementia syndromes and the MeDi because of its antioxidant properties [179]. A growing body of evidence implicated oxidative damage in the pathogenesis of AD, with increased presence of lipid peroxidation, nitration, free carbonyls, and nucleic acid oxidation [180, 181]. Higher oxidative stress has also been clearly invoked in MCI [182-184]. Complex phenols and many other substances with important antioxidant properties such as olive oil [185], wine, fruits and vegetables, vitamins C and E, and carotenoids [186-188] are found in high concentrations in the typical components of the MeDi [189]. Typical Mediterranean meals [190] or meals rich in typical Mediterranean food elements such as olive oil [191] or pomegranate juice [192] have been shown to increase enzymes with antioxidant properties such as paraoxonase and plasma carotenoids [190]. Food intervention studies with either different tomato products, a very commonly used food in Mediterranean areas [189], or a typical Mediterranean dish such as gazpacho, a cold vegetable soup that contains about 75% vegetables (tomato, cucumber, pepper), 2%-10% olive oil, and other minor components (onion, garlic, wine vinegar, and sea salt) [193] have indicated significant reductions of markers of oxidative stress such as isoprostanes. Therefore, the MeDi could be capturing the composite effect of dietary antioxidants and this could, at least partially, explain the association with a lower risk of dementia and predementia syndromes.

Finally, inflammation is another mechanism involved in the pathogenesis of dementia and predementia syndromes that is in general reduced with higher adherence to the MeDi [194, 195]. In fact, inflammation is another potential mechanism for AD pathogenesis [196, 197], has been found to be associated with a higher risk for AD, VaD, and cognitive

decline [18, 19], and links between MCI and higher inflammatory states have been also demonstrated [198-200]. In fact, C-reactive protein (CRP), an inflammatory marker that has been detected in SPs and NFTs in the brains of patients with AD [201, 202] and is up-regulated in AD brains [203, 204] and serum [205], has been proposed as a possible biomarker for AD [206], naMCI [207], and MCI [207, 208]. Higher adherence to the MeDi has been associated with lower CRP levels in both observational [170, 209, 210] and interventional [177, 190] studies. Furthermore, interleukin 6 (IL-6), a cytokine that mediates inflammatory reactions, has been consistently detected in diffuse early SPs without neuritic pathologic abnormalities of cortical regions of patients with AD [206] and has been associated with greater cognitive decline [211] and an increased risk of dementia during follow-up [212]. Tyrosol and caffeic acid, both found in extra virgin olive oil and in wine (which are essential components of the MeDi), have been shown to significantly reduce IL-6 production from peripheral blood mononuclear cells of healthy volunteers [213]. In the ATTICA study, subjects in the highest tertile of MeDi adherence have been reported to have 17% lower IL-6 serum levels [170], and in the Nurses' Health Study, there was a 16% reduction in IL-6 levels for subjects belonging to the upper quintile of a MeDi adherence index [210]. In a clinical trial, as compared with subjects assigned to the control diet, the subjects assigned to the MeDi had significantly reduced levels of IL-6 [177]. Higher adherence to the MeDi is in general associated with significant reduction in a series of other inflammatory markers, including white blood cell counts and others [170]. Thus, it is possible that the MeDi may, at least partially, lower the risk for dementia and predementia syndromes by affecting inflammatory processes. It is also possible that moderate lifestyles in general, which obviously vary according to different cultural environments, protect from cognitive impairment. Thus, it may not be the direct effect of MeDi or MeDi components (UFA, elevated fish consumption, moderate alcohol intake, and fruits and vegetables intake) or specific substances in MeDi components (antioxidant or anti-inflammatory compounds) that provide the protection, but the higher adherence to the MeDi may be an indicator of a complex set of favourable social and lifestyle factors [13].

Unsaturated Fatty Acids, Fish Consumption, and Cognitive Decline

As extensively seen above, the association between high adherence to the MeDi and lower risk for AD and cognitive decline may be mediated by the composite effect of some of its beneficial components. Some of the inconsistencies regarding some of the above dietary elements and risk for AD in the existing literature may be a result of failure to consider possible additive and interactive (antagonistic or synergistic) effects among nutritional components, which may be captured in a composite dietary pattern such as the MeDi [152]. In particular, different pathways could contribute to the neuroprotective as well as the neurotrophic properties of UFA [31, 33]. An underlying biological mechanism for the observed associations between UFA and dementia and predementia syndromes disease is currently unknown. In particular, several hypotheses could explain the protective effect of MUFA intake against ARCD and AD [31, 33]. In the older

subjects of the ILSA, the prolonged protection of MUFA intake against ARCD, may be linked to the relevant quota of antioxidant compounds in olive oil, including low molecular weight phenols [31, 33]. Moreover, substitution of caloric energy from SFA with UFA has been shown to lower low-density lipoprotein (LDL) cholesterol levels and increase high-density lipoprotein (HDL) cholesterol levels [214]. Other studies have found that high-fat/high-cholesterol diets increased and cholesterol-lowering drugs decreased A β peptide deposition and AD-related abnormalities [16]. Therefore, high MUFA intake may be a marker for other dietary factors responsible for protection against cognitive disorders, i.e. the great amount of tocopherol and polyphenols, the antioxidant compounds of olive oil, the low intake of SFA, and the not relevant quota of PUFA, in which n-3 are less represented than in Northern European population.

The neuroprotective effects of dietary UFA could rely on their impact on membrane architecture. In fact, UFA have an important role in maintaining the structural integrity of neuronal membranes, determining the fluidity of synaptosomal membranes and thereby regulating neuronal transmission. Furthermore, essential fatty acids can modify the activity of certain membrane-bound enzymes (phospholipase A2, protein kinase C, and acetyltransferase), and the function of the neurotransmitters' receptors. Moreover, free fatty acids, lipid metabolites, and phospholipids modify the function of membrane proteins including ion channels [31, 33]. Finally, fatty acid composition of neuronal membranes in advancing age demonstrated an increase in MUFA content and a decrease in PUFA content [31, 33].

In adult rats, learning and cognitive behaviour are related to brain docosahexaenoic acid (DHA) status, which, in turn, is related to the levels of the dietary n-3 PUFA [215]. In fact administration of DHA seems to improve learning ability in A β -infused rats [216] and inhibit decline in avoidance learning ability in the AD model rats, associated with an increase in the cortico-hippocampal n-3/n-6 ratio, and a decrease in neuronal apoptotic products [217]. In other transgenic AD mouse models, DHA also protects against dendritic pathology [218], prevents neuronal apoptosis induced by soluble A β peptides [219], increases synaptic protein and phospholipid densities [220], reduces the intraneuronal accumulation of both A β and tau protein [221], and inhibits degradative endopeptidase activities [222]. Some experimental evidence suggested that essential n-3 PUFA effectively lower A β production in transgenic mice as reported in studies from several laboratories [220, 223, 224].

There are several published studies on human infant subjects in which breastfeeding, which leads to higher DHA concentrations in the brain, or n-3 PUFA supplementation, is related to better cognitive performance at later age [31, 33]. The n-3 PUFA from fish may be inversely associated with dementia because it lowers the risk of thrombosis, stroke, cardiovascular disease, and cardiac arrhythmia, reducing the risk of thromboembolism in the brain and consequently of lacunar and large infarcts that can lead to VaD and AD [31, 33]. Furthermore, the n-3 PUFA may be important as lipids in the brain, particularly for the possible influence of DHA on the physical properties of the brain that are essential for its function [31, 33]. Furthermore, fish oil was a better

source than α -linolenic acid for the incorporation of n-3 PUFA into rat brain phospholipid subclasses [31, 33]. On the contrary, high linoleic acid intake (n-6 PUFA) may increase the susceptibility of LDL cholesterol to oxidation, which makes it more atherogenic, even if the association between linoleic acid and atherosclerosis is controversial [31, 33]. Therefore, the ratio of dietary n-3/n-6 PUFA intake may influence the potential role of PUFA on cognitive decline and dementia, and the optimal ratio of n-6/n-3 for a healthier diet should be $<5/1$ [31, 33]. Similar indications for cognitive decline or dementia, at present, are not available, but n-3 PUFA, particularly DHA status, should not be considered independently of n-6 PUFA intake [80]. Finally, a high dietary intake of SFA and cholesterol increases the risk for cardiovascular disease, and therefore for cognitive decline, VaD, and AD [31, 33]. A diet high in SFA and cholesterol may increase hippocampal accumulation of A β [224]. Diet high in fat/SFA may lead to cognitive impairment also through its effects on for example hyperinsulinemia [214], oxidative stress, or endothelial damage [31, 33]. In experimental studies SFA have decreased the levels of hippocampal brain-derived neurotrophic factor (BDNF) that are relevant for neuronal plasticity and the maintenance of cognitive functions [225]. On the contrary, treatment for four weeks with a Mediterranean-inspired diet rich in n-3 PUFA decreased blood lipids in healthy individuals with a low-risk profile for cardiovascular disease, with a beneficial effect also on vascular function and oxidative stress [226].

The inverse relationship between fish consumption and dementia could be explained by the biological hypothesis that the potentially protective active ingredients are the n-3 PUFA content in fish, with fatty fish containing considerably greater amounts of n-3 PUFA than white fish: approximately 2g and 0.3g n-3 PUFA per 100g average fish, respectively [227]. However, other constituents of fish could prevent dementia. For instance, fish is also a good source of selenium, a powerful antioxidant, and fatty fish traditionally consumed in Northern Europe are not only rich in n-3 PUFA, but also in vitamin D which is a recently recognized neurosteroid hormone [228, 229]. Vitamin D is essential for neurophysiological function, regulation of neurotransmitters or neurotrophic factors, and cerebral anti-oxidant, anti-inflammatory and anti-ischemic mechanisms [228, 229]. A recent systematic literature review suggested that vitamin D insufficiency may instead cause neuropathological dysfunction such as cognitive impairment [228]. These findings were subsequently confirmed in more than 750 older community-dwelling women with a 2-fold higher risk of cognitive impairment for vitamin D deficiency compared to normal status [229]. Therefore, the apparent protective effect of fish consumption against dementia could be explained by the impact of a health-promoting lifestyle or by the fish itself. In this perspective, the intake of n-3 PUFA, antioxidant compounds, and vitamin D should be taken into account.

Alcohol Intake, Wine Consumption, and Cognitive Decline

The mechanism by which low to moderate alcohol intake could be protective against cognitive decline impairment or decline in older age or against predementia and dementia syndromes is, at present, unclear. Given that alcohol drinking

influenced in particular measures of psychomotor speed, episodic memory, and executive function, it was supposed that WMLs would play a neuropathological role [105]. In fact, alcohol drinking has been associated with fewer brain infarcts and was shown to have a U shape relationship with WMLs [230]. WMLs and infarcts, in turn, may reflect a vascular mechanism responsible for the observed association between alcohol and cognitive functions. Furthermore, different mechanisms may underly the adverse effects of heavy drinking and the potential beneficial effects of low to moderate drinking. In fact, higher doses of alcohol may affect cognitive functioning through increased release of acetylcholine from the hippocampus [231], which was linked to problems with memory and attention [232]. On the other hand, animal evidence indicated that low doses of alcohol may stimulate the release of hippocampal acetylcholine [233], that regulates learning and memory. Finally, moderate alcohol consumption was associated with reduced cardiovascular risk factors [234] and lower rates of cardiovascular disease [235], which might protect brain vasculature and prevent subclinical strokes, resulting in better cognitive performance.

Alcohol consumption might protect from dementia by effects on the cerebral vasculature, supporting the observation that light to moderate alcohol intake might be protective against ischemic stroke [235]. Moreover, a light to moderate alcohol use is associated with a lower prevalence of MRI-defined WMLs and sub-clinical infarcts [236], although MRI abnormalities, HDL cholesterol levels, and fibrinogen levels only marginally influenced the association of alcohol consumption and dementia in the Cardiovascular Heart Study [237]. The suggested protection of low to moderate alcohol use against CVD may explain the possible protection also against VaD. In the Rotterdam study, the protective effect of alcohol consumption was found mainly for VaD, and the authors suggested that moderate alcohol intake might protect against dementia via a reduction in vascular risk factors [109]. In fact, moderate doses of alcohol may increase prostacyclin concentrations, reduce the generation of thromboxane A₂, and inhibit platelet function [37]. They may increase plasma levels of endogenous tissue-type plasminogen activator, a serine protease that regulates intravascular fibrinolysis, and fibrinolytic activity while decreasing plasma fibrinogen levels [37]. It is also known that alcohol is associated with increased levels of HDL cholesterol, its subfractions HDL2 and HDL3, and its associated apolipoproteins A-I and A-II [37]. The association with HDL cholesterol is deemed to account for up to a half of the reduction in coronary events associated with moderate alcohol consumption [35].

While the above-cited factors affected the risk of unspecified dementia and, probably, of VaD, other experimental and clinical findings may partly explain the suggested protection of low to moderate alcohol consumption against AD. In fact, small amounts of alcohol have been reported to be associated with a lower prevalence of vascular brain findings and, in APOE $\epsilon 4$ carriers, with hippocampal and amygdalar atrophy as assessed by MRI [236]. Experimental studies found that ethanol initially increases hippocampal acetylcholine release, which could conceivably improve memory performance [231].

Processes that originate, modulate, or precipitate the deposition of A β in the brain, such as oxidative stress, rather than vascular processes, may better explain the development of AD, and the vascular effects of the alcohol component of alcoholic beverages may not be enough to explain the protective effects of the moderate intake of alcohol from dementia. Wine consumption may exert a protective effect, either through alcohol intake itself, through the antioxidant effects of polyphenols richly represented in red wine [37], or through both. The latter effects, of course, are independent of alcohol and, in fact, have been also associated with alcohol-free red wine [238]. The constituents of red wine have also potentially beneficial vascular effects, enhance endothelial nitric oxide release, and reduce atherosclerosis in APOE-deficient mice [37]. Red-wine polyphenols are a complex mixture of flavonoids (mainly anthocyanins and flavan-3-ols) and non-flavonoids (such as resveratrol and gallic acid). Flavan-3-ols are the most abundant, with oligomeric and polymeric procyanidins (condensed tannins) often representing 25–50% of the total phenolic constituents [239]. Given the link between VaD, vascular risk, and the increasing body of evidence suggesting that AD may be influenced by vascular factors [11, 14], it may be concluded that the vascular protection associated to wine consumption decreases the risk of incident dementia/AD. In fact, in the 5-year follow-up *Personnes Agées QUID* (PAQUID) cohort, a significant inverse association between flavonoid intake and the risk of dementia was found [240]. It has been also suggested that the antioxidant properties of the flavonoids in wine may help prevent the oxidative damage implicated in dementia. In fact, oxidative stress may also develop in the brain, leading to neuronal death by various mechanisms such as formation of A β , DNA damage, mitochondrial dysfunction and abnormal tau protein [181, 182]. The presence in wine of nonalcoholic components, such as particular antioxidants, could explain a differential effect of wine on dementia. In fact, liquor has been shown to have less antioxidant activity than wine [241]. Nonetheless, in some studies on the neuroprotective role of moderate alcohol consumption, the most typically consumed alcohol types were beer and spirits [105, 119]. Finally, sirtuins, conserved proteins with a NAD⁺-dependent deacetylase and mono-ADP-ribosyltransferase activity, and particularly SIRT1 has been linked to the protective effects of red wine and calorie restriction in *in vivo* models of AD, and this had shed new light on the importance of moderate wine consumption and reduced food intake as AD-preventing behavioral strategies, suggesting a possible molecular mechanism underlying these AD risk modulators [242]. However, the beneficial effects of wine drinking and calorie restriction might involve molecular pathways other than sirtuins, and the bioavailability of resveratrol, the most potent natural SIRT1 activator, is quite low and would require more than a “moderate” wine consumption to reach next-to therapeutic concentrations [243]. In fact, even after a 25 mg oral dose, unconjugated resveratrol blood level was less than 5 ng/mL, and a resveratrol intake of 25 mg corresponds to the consumption of more than 4 liters of red wine (assuming a red wine mean resveratrol content of 6 mg/L) [244].

Fruits and Vegetables Intake and Cognitive Decline

Fruits and vegetables reflect distinct food groups in the MeDi and in the American Diabetes Association food pyra-

mid, and the benefits of these food groups could be provided through the anti-oxidant properties that foods in this class provide [245]. In fact, fruits and vegetables are rich sources of antioxidant nutrients and bioactive compounds, such as flavonoids, that are thought to be involved in neurodegenerative processes. This is in accordance with previous finding of a protective effect of higher consumption of dietary flavonoids against dementia [240] and cognitive decline [246] in the PAQUID study. Furthermore, there is growing evidence indicating that oxidative damage caused by the A β peptide in the pathogenesis of AD may be hydrogen peroxide (H₂O₂)-mediated [247]. Recent studies have shown that flavonols, a class of flavonoids, from apple and citrus juices, such as quercetin, are able to cross the blood-brain barrier [248] and show neuroprotection against H₂O₂ [249] in human neuroblastoma cells, which may or may not behave like neurons in the intact brain, preventing also A β -induced calcium influx and apoptosis, each of which results in part due to increased ROS. The effect of flavonols from citrus is similar to vitamin C, but quercetin from apple juice confers stronger neuroprotection than vitamin C [250]. Early *in vitro* studies reported that polyphenol flavonols, such as quercetin, protect mammalian and bacterial cells from toxicity induced by H₂O₂, but that α -tocopherol is not effective [251]. Recent *in vitro* studies show that many polyphenols, including flavonols (eg, quercetin and others), protect mouse hippocampal cells from oxidative glutamate and H₂O₂ toxicity [249]. Additional animal studies showed that chronic administration of flavonols or apple juice protects against aging and cognitive impairment induced by lipopolysaccharide, genetic, and dietary vitamin deficiency in animal models [252, 253]. However, the Nurses'Health study [145] and the CHAP [146] found a protective effect of higher vegetable consumption, but not higher fruit consumption, against cognitive decline. A partial explanation of this finding could be that vegetables contain more of vitamin E than do fruits, and this antioxidant nutrient was previously shown to be inversely related to risk of cognitive decline in the CHAP study population [254]. This explanation is supported by the fact that green leafy vegetables (which generally contain more vitamin E than other vegetable types) had the strongest inverse relation with cognitive decline in the CHAP [146]. Vegetables, but not fruits, are also typically consumed with added fats (e.g., salad dressings, mayonnaise, margarine or butter), and fats increase the absorption of vitamin E and other fat soluble antioxidant nutrients, such as carotenoids and flavonoids [146].

Other possible mechanisms linking fruits and vegetables intake with neuroprotection could be of vascular origin. In fact, a meta-analysis showed that regular consumption of fruits and vegetables decreased the risk for stroke [255], which could explain their protective effect against the vascular component of dementia. In addition to anti-oxidant and vascular mechanisms, anti-inflammatory mechanisms have been invoked as contributing to the benefit of these food groups. Studies have demonstrated that dietary levels of fruit intake and of vitamin C are inversely related to levels of CRP [256], and vegetable intake is inversely associated with tissue plasminogen activator, a marker of endothelial dysfunction [245]. Furthermore, aside from their antioxidant properties, many polyphenols, such as quercetin, have potent

anti-inflammatory properties [257]. Recent clinical trials have shown that intake of orange juice significantly reduces plasma concentrations of F2-isoprostanes, a valid biomarker of oxidative stress [258]. In addition to antioxidant compounds, fruit and vegetable juices also may possess other protective components, such as folate and minerals [259], and a high serum level of folate was found to be associated with a decreased risk of AD [260].

CONCLUSIONS

Given the very limited therapeutic value of drugs currently used in the treatment of cognitive impairment and dementia, it results evident the necessity to potentially individualize new strategies able to prevent and to slow down the progression of predementia and dementia syndromes. Only recently higher adherence to a Mediterranean-type diet was associated with cognitive decline although the MeDi combines several foods and nutrients already separately proposed as potential protective factors against dementia and predementia syndromes. In fact, at present, epidemiological evidence suggested a possible association among fish consumption, MUFA, and PUFA (particularly, n-3 PUFA) and reduced risk of cognitive decline and dementia. Furthermore, light to moderate alcohol use may be associated with a reduced risk of incident dementia and AD, while for VaD, cognitive decline, and predementia syndromes the current evidence is only suggestive of a protective effect. Finally, some lines of evidence suggested older healthy subjects consuming a diet rich in fruits and vegetables have higher plasma levels of lipophilic antioxidant micronutrients, lower levels of biomarkers related to oxidative stress, and better scores on neuropsychological evaluation compared to subjects with low intakes of fruits and vegetables. On the basis of this evidence, the evaluation of the MeDi on cognitive outcomes seems of particular interest and recent prospective studies focused on dementia and predementia syndromes appeared to be really promising. In fact, higher adherence to a Mediterranean-type diet has been associated with slower cognitive decline, with a trend towards a reduced risk for developing MCI, with reduced risk of progression from MCI to AD, decreased risk for AD, and reduced all-causes mortality in AD patients. These findings suggested that adherence to the MeDi may affect not only the risk for AD, but also for predementia syndromes, probably influencing the evolution of cognitive performances over time and subsequent disease progression. Future studies on this topic should probably address different constructs of predementia syndromes and subtypes of MCI, also accounting for the possible role of APOE ϵ 4 allele, a major genetic risk factor for AD and cognitive decline, in modifying the possible associations between the MeDi and cognitive decline. Further research is also needed to allow the generalization of these results on other populations and to propose the MeDi as a potential preventive approach against dementia and cognitive decline.

However, while for vascular risk factors there was a clear evidence about a protective role of MeDi and its nutrients in preventing all cardiovascular conditions linked to dementia, for the other possible mechanisms it was only possible to suggest hypothetical biological pathways taken into account the results from animal studies. Therefore, the lack of reproducibility in some results and the speculative aspect of the

biological pathways presented, suggested us to be cautious. However, although it was difficult to define the real impact of these biological findings, these studies provided other possible explanation of the neuroprotective effect of MeDi and its micro- and macronutrients that is not only linked to inflammatory and oxidative-related mechanisms. In fact, the efficacy of nutritionally derived compounds as neuroprotective agents is increasingly supported by empirical evidence. Plant-derived molecules including polyphenols have demonstrated neuroprotective activities in cell culture and animal models. However, the molecular mechanisms underlying their protective effects were generally not completely understood and further investigations will be essential to provide links between MeDi nutrients and brain function in a mechanistic, dynamic and quantitative way. Recommendations regarding future research on the effects of the adherence to the MeDi on predementia and dementia syndromes include properly designed randomized clinical trials (RCTs) that are sufficiently powered and with an adequate length, although this approach could have some potential limitations. In fact, given the long preclinical phase of dementia, it may be difficult to execute appropriate RCTs over a long enough time and with large enough samples to draw accurate and repeatable conclusions. Based in part on the evidence discussed in this review article, it appears possible that in the near future we might consider employing dietary intervention in preventative and possibly therapeutic strategies for dementia and predementia syndromes. Nonetheless, at present, no definitive dietary recommendations are possible. However, high levels of consumption of fats from fish, vegetable oils, non-starchy vegetables, low glycemic fruits, and diet low in foods with added sugars and with moderate wine intake should be encouraged. In fact, this dietary advice provides a variety of antioxidant nutrients, is in accordance with recommendations for lowering the risk of cardiovascular disease, obesity, diabetes, and hypertension and might open new routes for the prevention and management of cognitive decline and dementia to supplement existing symptomatic approaches.

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REFERENCES

- [1] Lobo A, Launer LJ, Fratiglioni L, Andersen K, Di Carlo A, Breteler MM, *et al.*, for the Neurologic Diseases in the Elderly Research Group. Prevalence of dementia and major subtypes in Europe. A collaborative study of population based-cohorts. *Neurology* 54: S4-S9 (2000).
- [2] Qiu C, De Ronchi D, Fratiglioni L. The epidemiology of the dementias: an update. *Curr Opin Psychiatry* 20: 380-385 (2007).
- [3] Walter J, Kaether C, Steiner H, Haass C. The cell biology of Alzheimer's disease: uncovering the secrets of secretases. *Curr Opin Neurobiol* 11: 585-590 (2001).
- [4] Hardy J, Allsop D. Amyloid deposition as the central event in the aetiology of Alzheimer's disease. *Trends Pharmacol Sci* 12: 383-388 (1991).
- [5] Panza F, D'Introno A, Colacicco AM, Capurso C, Del Parigi A, Caselli RJ, *et al.* Current epidemiology of mild cognitive impairment and other predementia syndromes. *Am J Geriatr Psychiatry* 13: 633-644 (2005).

- [6] Winblad B, Palmer K, Kivipelto M, Jelic V, Fratiglioni L, Wahlund LO, *et al.* Mild cognitive impairment—beyond controversies, towards a consensus: report of the International Working Group on Mild Cognitive Impairment. *J Intern Med* 256: 240-246 (2004).
- [7] Panza F, Frisardi V, Capurso C, D'Introno A, Colacicco AM, Imbimbo BP, *et al.* Late-life depression, mild cognitive impairment, and dementia: possible continuum? *Am J Geriatr Psychiatry* 18: 98-116 (2010).
- [8] American Psychiatric Association Committee on Nomenclature and Statistics. *Diagnostic and Statistical Manual of Mental Disorders*. American Psychiatric Association; Washington; DC (1994).
- [9] Levy R. Aging-associated cognitive decline. Working Party of the International Psychogeriatric Association in collaboration with the World Health Organization. *Int Psychogeriatrics* 6: 63-68 (1994).
- [10] Frisardi V, Solfrizzi V, Imbimbo BP, Capurso C, D'Introno A, Colacicco AM, *et al.* Towards Disease-Modifying Treatment of Alzheimer's Disease: Drugs Targeting β -Amyloid. *Curr Alzheimer Res* 7: 40-55 (2010).
- [11] Panza F, D'Introno A, Colacicco AM, Capurso C, Del Parigi A, Capurso SA, *et al.* Cognitive frailty: Predementia syndrome and vascular risk factors. *Neurobiol Aging* 27: 933-940 (2006).
- [12] Panza F, Solfrizzi V, Colacicco AM, D'Introno A, Capurso C, Torres F, *et al.* Mediterranean diet and cognitive decline. *Public Health Nutr* 7: 959-963 (2004).
- [13] Solfrizzi V, Capurso C, D'Introno A, Colacicco AM, Santamato A, Ranieri M, *et al.* Lifestyle-related factors in predementia and dementia syndromes. *Expert Rev Neurother* 8: 133-158 (2008).
- [14] Frisardi V, Solfrizzi V, Capurso C, Kehoe PG, Imbimbo BP, Santamato A, *et al.* Aluminum in the diet and Alzheimer's disease: from current epidemiology to possible disease-modifying treatment. *J Alzheimers Dis* 20: 17-30 (2010).
- [15] Qiu C, Winblad B, Fratiglioni L. The age-dependent relation of blood pressure to cognitive function and dementia. *Lancet Neurol* 4: 487-499 (2005).
- [16] Panza F, D'Introno A, Colacicco AM, Capurso C, Pichichero G, Capurso SA, Capurso A, Solfrizzi V. Lipid metabolism in cognitive decline and dementia. *Brain Res Rev* 51: 275-292 (2006).
- [17] Luchsinger JA, Gustafson DR. Adiposity, type 2 diabetes, and Alzheimer's disease. *J Alzheimers Dis* 16: 693-704 (2009).
- [18] Solfrizzi V, Scafato E, Capurso C, D'Introno A, Colacicco AM, Frisardi V, *et al.* Italian Longitudinal Study on Aging Working Group Metabolic syndrome, mild cognitive impairment, and progression to dementia. The Italian Longitudinal Study on Aging. *Neurobiol Aging*; doi:10.1016/j.neurobiolaging.2009.12.012 (2009).
- [19] Solfrizzi V, Scafato E, Capurso C, D'Introno A, Colacicco AM, Frisardi V, *et al.* Italian Longitudinal Study on Aging Working Group. Metabolic syndrome and the risk of vascular dementia. The Italian Longitudinal Study on Aging. *J Neurol Neurosurg Psychiatry* 81: 433-440 (2010).
- [20] Peters R. The prevention of dementia. *Int J Geriatr Psychiatry* 24: 452-458 (2009).
- [21] Kramer AF, Hahn S, Cohen NJ, Banich MT, McAuley E, Harrison CR, Chason J, Vakil E, Bardell L, Boileau RA, Colcombe A. Ageing, fitness and neurocognitive function. *Nature* 400: 418-419 (1999).
- [22] Ngandu T, von Strauss E, Helkala EL, Winblad B, Nissinen A, Tuomilehto J, Soininen H, Kivipelto M. Education and dementia what lies beneath the association? *Neurology* 69: 1442-1450 (2007).
- [23] Solfrizzi V, Panza F, Capurso A. The role of diet in cognitive decline. *J Neural Transm* 110: 95-110 (2003).
- [24] Grant WB. Dietary links to Alzheimer's disease. *Alzheimers Dis Rev* 2: 42-55 (1997).
- [25] Grant WB. Dietary links to Alzheimer's disease: 1999 Update. *J Alzheimers Dis* 1: 197-201 (1999).
- [26] Solfrizzi V, Colacicco AM, D'Introno A, Capurso C, Del Parigi A, Capurso SA, *et al.* Macronutrients; aluminium from drinking water and foods; and other metals in cognitive decline and dementia. *J Alzheimers Dis* 10: 303-330 (2006).
- [27] Luchsinger JA, Mayeux R. Dietary factors and Alzheimer's disease. *Lancet Neurol* 3: 579-587 (2004).
- [28] Luchsinger JA, Noble JM, Scarmeas N. Diet and Alzheimer's disease. *Curr Neurol Neurosci Rep* 7: 366-372 (2007).
- [29] Coley N, Andrieu S, Gardette V, Gillette-Guyonnet S, Sanz C, Vellas B, *et al.* Dementia prevention: methodological explanations for inconsistent results. *Epidemiol Rev* 30: 35-66 (2008).
- [30] Panza F, D'Introno A, Colacicco AM, Basile AM, Capurso C, Kehoe PG, *et al.* Vascular risk and genetics of sporadic late-onset Alzheimer's disease. *J Neural Transm* 111: 69-89 (2004).
- [31] Solfrizzi V, D'Introno A, Colacicco AM, Capurso C, Del Parigi A, Capurso S, *et al.* Dietary fatty acids intake: possible role in cognitive decline and dementia. *Exp Gerontol* 40: 257-270 (2005).
- [32] Assisi A, Banzi R, Buonocore C, Capasso F, Di Muzio V, Michelacci F, *et al.* Fish oil and mental health. the role of n-3 long-chain polyunsaturated fatty acids in cognitive development and neurological disorders. *Int Clin Psychopharmacol* 21: 319-336 (2006).
- [33] Solfrizzi V, Frisardi V, Capurso C, D'Introno A, Colacicco AM, Vendemiale G, *et al.* Dietary fatty acids in dementia and predementia syndromes: Epidemiological evidence and possible underlying mechanisms. *Ageing Res Rev* 9: 184-99 (2010).
- [34] Dai Q, Borenstein A, Wu Y, Jackson JC, Larson EB. Fruit and vegetable juices and Alzheimer's disease. the Kame project. *Am J Med* 119: 751-759 (2006).
- [35] Parrott MD, Greenwood CE. Dietary influences on cognitive function with aging. From high-fat diets to healthful eating. *Ann N Y Acad Sci* 1114: 389-397 (2007).
- [36] Peters R, Peters J, Warner J, Beckett N, Bulpitt C. Alcohol, dementia and cognitive decline in the elderly. a systematic review. *Age Ageing* 37: 505-512 (2008).
- [37] Panza F, Capurso C, D'Introno A, Colacicco AM, Frisardi V, Lorusso M, *et al.* Alcohol drinking, cognitive functions in older age, predementia, and dementia syndromes. *J Alzheimers Dis* 17: 7-31 (2009).
- [38] US Department of Health and Human Services, US Department of Agriculture. *Dietary guidelines for Americans 2005*. 6th ed. Washington, DC: US Government Printing Office (2005) (available at: www.healthierus.gov/dietaryguidelines).
- [39] Keys A. Seven countries: a multivariate analysis of death and coronary heart disease. Cambridge, MA: Harvard University Press (1980).
- [40] Serra-Majem L, Roman B, Estruch R. Scientific evidence of interventions using the Mediterranean diet: a systematic review. *Nutr Rev* 64: S27-47 (2006).
- [41] De Lorgeril M, Salen P, Martin J, Monjaud I, Delaye J, Marmelle N. Mediterranean diet, traditional risk factors and the rate of cardiovascular complications after myocardial infarction: final report of the Lyon diet heart study. *Circulation* 99: 799-785 (1999).
- [42] Willett WC, Sacks F, Trichopoulos A, Drescher G, Ferro-Luzzi A, Helsing E, *et al.* Mediterranean diet pyramid: a cultural model for healthy eating. *Am J Clin Nutr* 61: 1402S-1406S (1995).
- [43] Sofi F. The Mediterranean diet revisited: evidence of its effectiveness grows. *Curr Opin Cardiol* 24: 442-446 (2009).
- [44] Gomez-Pinilla F. Brain foods: the effects of nutrients on brain function. *Nat Rev Neurosci* 9: 568-578 (2008).
- [45] Pradignac A, Schlienger JL, Velten M, Mejean L. Relationships between macronutrient intake, handicaps, and cognitive impairments in free living elderly people. *Ageing Clin Exp Res* 7: 67-74 (1995).
- [46] Ortega RM, Requejo AM, Andres P, Lopez-Sobaler AM, Quintas ME, Redondo MR, *et al.* Dietary intake and cognitive function in a group of elderly people. *Am J Clin Nutr* 66: 803-809 (1997).
- [47] Solfrizzi V, Panza F, Torres F, Mastroianni F, Del Parigi A, Venezia A, *et al.* High monounsaturated fatty acids intake protects against age-related cognitive decline. *Neurology* 52: 1563-1569 (1999).
- [48] Manzato E, Roselli della Rovere G, Zambon S, Romanato G, Corti MC, Sartori L, *et al.* Cognitive functions are not affected by dietary fatty acids in elderly in the Pro.V.A. study population. *Ageing Clin Exp Res* 15: 83-86 (2003).
- [49] Kalmijn S, Feskens EJ, Launer LJ, Kromhout D. Polyunsaturated fatty acids, antioxidants, and cognitive functions in very old men. *Am J Epidemiol* 145: 33-41 (1997).
- [50] Hende B, Ducimitiere P, Berr C. Cognitive decline and fatty acid composition of erythrocyte membranes – The EVA Study. *Am J Clin Nutr* 77: 803-808 (2003).
- [51] Morris MC, Evans DA, Bienias JL, Tangney CC, Bennett DA, Wilson RS. Dietary fat intake and 6-year cognitive change in an older biracial community population. *Neurology* 62: 1573-1579 (2004).

- [52] Kalmijn S, van Boxtel MP, Ocke M, Verschuren WM, Kromhout D, Launer LJ. Dietary intake of fatty acids and fish in relation to cognitive performance at middle age. *Neurology* 62: 275-280 (2004).
- [53] Morris MC, Evans DA, Tangney CC, Bienias JL, Julie A, Schneider JA, *et al.* Dietary copper and high saturated and trans fat intakes associated with cognitive decline. *Arch Neurol* 63: 1085-1088 (2005).
- [54] Solfrizzi V, Colacicco AM, D'Introno A, Capurso C, Torres F, Rizzo C, *et al.* Dietary intake of unsaturated fatty acids and age-related cognitive decline: A 8.5-year follow-up of the Italian Longitudinal Study on Aging. *Neurobiol Aging* 27: 1694-1704 (2006).
- [55] Solfrizzi V, Colacicco AM, D'Introno A, Capurso C, Del Parigi A, Capurso SA, *et al.* Dietary polyunsaturated fatty acids intakes and rate of mild cognitive impairment. The Italian Longitudinal Study on Aging. *Exp Gerontol* 41: 619-627 (2006).
- [56] Psaltopoulou T, Kyrozis A, Stathopoulos P, Trichopoulos D, Vassilopoulos D, Trichopolou A. Diet, physical activity and cognitive impairment among elders: The EPIC-Greece cohort (European Prospective Investigation into Cancer and Nutrition). *Public Health Nutr* 11: 1054-1062 (2008).
- [57] Eskelinen MH, Ngandu T, Helkala EL, Tuomilehto J, Nissinen A, Soininen H, *et al.* Fat intake at midlife and cognitive impairment later in life: a population-based CAIDE study. *Int J Geriatr Psychiatry* 23: 741-747 (2008).
- [58] Vercambre MN, Boutron-Ruault MC, Ritchie K, Clavel-Chapelon F, Berr C. Long-term association of food and nutrient intakes with cognitive and functional decline: a 13-year follow-up study of elderly French women. *Br J Nutr* 102, 419-427 (2009).
- [59] Berr C, Portet F, Carriere I, Akbaraly TN, Fear C, Gourlet V, *et al.* Olive oil and cognition: results from the three-city study. *Dement Geriatr Cogn Disord* 28: 357-364 (2009).
- [60] Kalmijn S, Launer LJ, Ott A, Witteman JC, Hofman A, Breteler MM. Dietary fat intake and the risk of incident dementia in the Rotterdam study. *Ann Neurol* 42: 776-782 (1997).
- [61] Engelhart MJ, Geerlings MI, Ruitenberg A, Van Swieten JC, Hofman A, Witteman JC, *et al.* Diet and risk of dementia: Does fat matter? The Rotterdam Study. *Neurology* 59: 1915-1921 (2002).
- [62] Luchsinger JA, Tang MX, Shea S, Mayeux R. Caloric intake and the risk of Alzheimer disease. *Arch Neurol* 59: 1256-1263 (2002).
- [63] Barberger-Gateau P, Letenneur L, Deschamps V, Peres K, Dartigues JF, Renaud S. Fish, meat, and risk of dementia: cohort study. *BMJ* 325: 932-933 (2002).
- [64] Morris MC, Evans DA, Bienias JL, Tangney CC, Bennett DA, Aggarwal N, *et al.* Dietary fats and the risk of incident Alzheimer disease. *Arch Neurol* 60: 194-200 (2003).
- [65] Morris MC, Evans DA, Bienias JL, Tangney CC, Bennett DA, Wilson RS, *et al.* Consumption of fish and n-3 fatty acids and risk of incident Alzheimer disease. *Arch Neurol* 60: 940-946 (2003).
- [66] Laitinen MH, Ngandu T, Rovio S, Helkala E-L, Uusitalo U, Viitanen M, *et al.* Fat intake at midlife and risk of dementia and Alzheimer's disease: a population-based study. *Dement Geriatr Cogn Disord* 22: 99-107 (2006).
- [67] Barberger-Gateau P, Raffaitin C, Letenneur L, Berr C, Tzourio C, Dartigues JF, *et al.* Dietary patterns and risk of dementia: the Three-City cohort study. *Neurology* 69: 1921-1930 (2007).
- [68] Dangour AD, Allen E, Elbourne D, Fletcher A, Richards M, Uauy R. Fish Consumption and Cognitive Function among Older People in the UK: Baseline Data from the Opal Study. *J Nutr Health Aging* 13: 198-202 (2009).
- [69] Morris MC, Evans DA, Tangney CC, Bienias JL, Wilson RS. Fish consumption and cognitive decline with age in a large community study. *Arch Neurol* 62: 1849-1853 (2005).
- [70] Nurk E, Drevon CA, Refsum H, Solvoll K, Vollset SE, Nygård O, *et al.* Cognitive performance among the elderly and dietary fish intake: the Hordaland Health Study. *Am J Clin Nutr* 86: 1470-1478 (2007).
- [71] van Gelder BM, Tijhuis M, Kalmijn S, Kromhout D. Fish consumption, n-3 fatty acids, and subsequent 5-y cognitive decline in elderly men: the Zutphen Elderly Study. *Am J Clin Nutr* 85: 1142-1147 (2007).
- [72] Huang TL, Zandi PP, Tucker KL, Fitzpatrick AL, Kuller LH, Fried LP, *et al.* Benefits of fatty fish on dementia risk are stronger for those without APOE epsilon4. *Neurology* 65: 1409-1414 (2005).
- [73] Panza F, Frisardi V, Capurso C, D'Introno A, Colacicco AM, Di Palo A, *et al.* Polyunsaturated fatty acid and S-adenosylmethionine supplementation in predementia syndromes and Alzheimer's disease: a review. *TheScientificWorldJournal* 9: 373-389 (2009).
- [74] Freund-Levi Y, Eriksdottir-Jönhagen M, Cederholm T, Basun H, Faxén-Irving G, Garlind A, *et al.* ω -3 fatty acid treatment in 174 patients with mild to moderate Alzheimer disease: OmegaAD Study: a randomized double-blind trial. *Arch Neurol* 63: 1402-1408 (2009).
- [75] Kotani S, Sakaguchi E, Warashina S, Matsukawa N, Ishikura Y, Kiso Y, *et al.* Dietary supplementation of arachidonic and docosahexaenoic acids improves cognitive dysfunction. *Neurosci Res* 56: 159-164 (2006).
- [76] Chiu CC, Su KP, Cheng TC, Liu HC, Chang CJ, Dewey ME, *et al.* The effects of omega-3 fatty acids monotherapy in Alzheimer's disease and mild cognitive impairment: a preliminary randomized double-blind placebo-controlled study. *Prog Neuropsychopharmacol Biol Psychiatry* 32: 1538-1544 (2008).
- [77] Freund-Levi Y, Basun H, Cederholm T, Faxén-Irving G, Garlind A, Grut M, *et al.* Omega-3 supplementation in mild to moderate Alzheimer's disease: effects on neuropsychiatric symptoms. *Int J Geriatr Psychiatry* 23: 161-169 (2008).
- [78] van de Rest O, Geleijnse JM, Kok FJ, van Staveren WA, Hoefnagels WH, Beekman AT, *et al.* Effect of fish-oil supplementation on mental well-being in older subjects: a randomized, double-blind, placebo-controlled trial. *Am J Clin Nutr* 88: 706-713 (2008).
- [79] van de Rest O, Geleijnse JM, Kok FJ, van Staveren WA, Dullemeijer C, Oudejans MG, *et al.* Effect of fish oil on cognitive performance in older subjects: a randomized, controlled trial. *Neurology* 71: 430-438 (2008).
- [80] Cunnane SC, Plourde M, Pifferi F, Bégin M, Féart C, Barberger-Gateau P. Fish, docosahexaenoic acid and Alzheimer's disease. *Prog Lipid Res*; doi:10.1016/j.plipres.2009.04.001 (2009).
- [81] Friedland RP. Fish consumption and the risk of Alzheimer disease: is it time to make dietary recommendations? *Arch Neurol* 60: 923-924 (2003).
- [82] Solfrizzi V, Frisardi V, Capurso C, D'Introno A, Colacicco AM, Vendemmiale G, *et al.* Moderate alcohol consumption, apolipoprotein E, and neuroprotection. *Arch Neurol* 66: 541-542 (2009).
- [83] Goodwin JS, Sanchez CJ, Thomas P, Hunt C, Garry PJ, Goodwin JM. Alcohol intake in a healthy elderly population. *Am J Public Health* 77: 173-177 (1987).
- [84] Christian JC, Reed T, Carmelli D, Page WF, Norton JA Jr, Breitner JC. Self-reported alcohol intake and cognition in aging twins. *J Stud Alcohol* 56: 414-416 (1995).
- [85] Hendrie HC, Gao S, Hall KS, Hui SL, Unverzagt FW. The relationship between alcohol consumption, cognitive performance, and daily functioning in an urban sample of older Black Americans. *J Am Geriatr Soc* 44: 1158-1165 (1996).
- [86] Launer LJ, Feskens EJ, Kalmijn S, Kromhout D. Smoking; drinking; and thinking. The Zutphen Elderly Study. *Am J Epidemiol* 143: 219-227 (1996).
- [87] Elias PK, Elias MF, D'Agostino RB, Silbershatz H, Wolf PA. Alcohol consumption and cognitive performance in the Framingham Heart Study. *Am J Epidemiol* 150: 580-589 (1999).
- [88] Carmelli D, Swan GE, Reed T, Schellenberg GD, Christian JC. The effect of apolipoprotein E epsilon4 in the relationships of smoking and drinking to cognitive function. *Neuroepidemiology* 18: 125-133 (1999).
- [89] Bond GE, Burr R, McCurry SM, Graves AB, Larson EB. Alcohol, aging, and cognitive performance in a cohort of Japanese Americans aged 65 and older: the Kame Project. *Int Psychogeriatr* 13: 207-223 (2001).
- [90] Bond GE, Burr R, Rice MM, McCurry SM, Graves AB, Teri L, *et al.* Alcohol, aging, and cognitive performance: a cross-cultural comparison. *J Aging Health* 15: 371-390 (2003).
- [91] Dufouil C, Ducimetière P, Alperovitch A. Sex differences in the association between alcohol consumption and cognitive performance. EVA Study Group. *Epidemiology of Vascular Aging*. *Am J Epidemiol* 146: 405-412 (1997).
- [92] Schinka JA, Vanderploeg RD, Rogish M, Graves AB, Mortimer JA, Ordoric PI. Effects of the use of alcohol and cigarettes on cognition in elderly adults. *J Int Neuropsychol Soc* 8: 811-818 (2002).
- [93] Verhaegen P, Borchelt M, Smith J. Relation between cardiovascular and metabolic disease and cognition in very old age: cross-sectional and longitudinal findings from the Berlin aging study. *Health Psychol* 22: 559-569 (2003).

- [94] Lindeman RD, Wayne SJ, Baumgartner RN, Garry PJ. Cognitive Function in Drinkers Compared to Abstainers in The New Mexico Elder Health Survey. *J Gerontol A Biol Sci Med Sci* 60: 1065-1070 (2005).
- [95] Reid MC, Van Ness PH, Hawkins KA, Towle V, Concato J, Guo Z. Light to moderate alcohol consumption is associated with better cognitive function among older male veterans receiving primary care. *J Geriatr Psychiatry Neurol* 19: 98-105 (2006).
- [96] Yoshitake T, Kiyohara Y, Kato I, Ohmura T, Iwamoto H, Nakayama K, *et al.* Incidence and risk factors of vascular dementia and Alzheimer's disease in a defined elderly Japanese population: the Hisayama Study. *Neurology* 45: 1161-1168 (1995).
- [97] Dufouil C, Tzourio C, Brayne C, Berr C, Amouyel P, Alperovitch A. Influence of apolipoprotein E genotype on the risk of cognitive deterioration in moderate drinkers and smokers. *Epidemiology* 11: 280-284 (2000).
- [98] Leibovici D, Ritchie K, Ledesert B, Touchon J. The effects of wine and tobacco consumption on cognitive performance in the elderly: a longitudinal study of relative risk. *Int J Epidemiol* 28: 77-81 (1999).
- [99] Stampfer MJ, Kang JH, Chen J, Cherry R, Grodstein F. Effects of moderate alcohol consumption on cognitive function in women. *N Engl J Med* 352: 245-253 (2005).
- [100] Hébert LE, Taylor JO, Evans DA. Relation of smoking and low-to-moderate alcohol consumption to change in cognitive function: a longitudinal study in a defined community of older persons. *Am J Epidemiol* 137: 881-891 (1993).
- [101] Launer LJ, Feskens EJ, Kalmijn S, Kromhout D. Smoking, drinking, and thinking. The Zutphen Elderly Study. *Am J Epidemiol* 143: 219-227 (1996).
- [102] Broe GA, Creasey H, Jorm AF, Bennett HP, Casey B, Waite LM, *et al.* Health habits and risk of cognitive impairment and dementia in old age: a prospective study on the effects of exercise, smoking and alcohol consumption. *Aust N Z J Public Health* 22: 621-623 (1998).
- [103] Cervilla JA, Prince M, Mann A. Smoking, drinking, and incident cognitive impairment: a cohort community based study included in the Gospel Oak Project. *J Neurol Neurosurg Psychiatry* 68: 622-626 (2000).
- [104] Galanis DJ, Joseph C, Masaki KH, Petrovich H, Ross GW, White L. A longitudinal study of drinking and cognitive performance in elderly Japanese American men: the Honolulu-Asia aging study. *Am J Public Health* 90: 1254-1259 (2000).
- [105] Ngandu T, Helkala EL, Soininen H, Winblad B, Tuomilehto J, Nissinen A, *et al.* Alcohol drinking and cognitive functions: findings from the Cardiovascular Risk Factors Aging and Dementia (CAIDE) Study. *Dement Geriatr Cogn Disord* 23: 140-149 (2007).
- [106] Yoshitake T, Kiyohara Y, Kato I, Ohmura T, Iwamoto H, Nakayama K, *et al.* Incidence and risk factors of vascular dementia and Alzheimer's disease in a defined elderly Japanese population: the Hisayama Study. *Neurology* 45: 1161-1168 (1995).
- [107] Orgogozo JM, Dartigues JF, Lafont S, Letenneur L, Commenges D, Salamon R, *et al.* Wine consumption and dementia in the elderly: a prospective community study in the Bordeaux area. *Rev Neurol (Paris)* 153: 185-192 (1997).
- [108] Lemeshow S, Letenneur L, Dartigues JF, Lafont S, Orgogozo JM, Commenges D. Illustration of analysis taking into account complex survey considerations: the association between wine consumption and dementia in the PAQUID Study. *Am J Epidemiol* 148: 298-306 (1998).
- [109] Ruitenberg A, van Swieten JC, Witteman JC, Mehta KM, van Duijn CM, Hofman A, *et al.* Alcohol consumption and risk of dementia: the Rotterdam Study. *Lancet* 359: 281-286 (2002).
- [110] Lindsay J, Laurin D, Verreault R, Hébert R, Helliwell B, Hill GB, *et al.* Risk factors for Alzheimer's disease: a prospective analysis from the Canadian Study of Health and Aging. *Am J Epidemiol* 156: 445-453 (2002).
- [111] Huang W, Qiu C, Winblad B, Fratiglioni L. Alcohol consumption and incidence of dementia in a community sample aged 75 years and older. *J Clin Epidemiol* 55: 959-964 (2002).
- [112] Deng J, Zhou DH, Li J, Wang YJ, Gao C, Chen M. A 2-year follow-up study of alcohol consumption and risk of dementia. *Clin Neurol Neurosurg* 108: 378-383 (2006).
- [113] Kivipelto M, Helkala EL, Hänninen T, Laakso MP, Hallikainen M, Alhainen K, *et al.* Midlife vascular risk factors and Alzheimer's disease in later life: longitudinal; population based study. *BMJ* 322: 1447-1451 (2001).
- [114] Fujishima M, Kiyohara Y. Incidence and risk factors of dementia in a defined elderly Japanese population the Hisayama study. *Ann N Y Acad Sci* 977: 1-8 (2002).
- [115] Järvenpää T, Rinne JO, Koskenvuo M, Riihinen I, Kaprio J. Binge drinking in midlife and dementia risk. *Epidemiology* 16: 766-771 (2005).
- [116] Simons LA, Simons J, McCallum J, Friedlander Y. Lifestyle factors and risk of dementia: Dubbo study of the elderly. *Med J Aust* 184: 68-70 (2006).
- [117] Mehlig K, Skoog I, Guo X, Schütze M, Gustafson D, Waern M, *et al.* Alcoholic beverages and incidence of dementia: 34-year follow-up of the prospective population study of women in Goteborg. *Am J Epidemiol* 167: 684-691 (2008).
- [118] Solfrizzi V, D'Introno A, Colacicco AM, Capurso C, Del Parigi A, Baldassarre G, *et al.* Italian Longitudinal Study on Aging Working Group. Alcohol consumption; mild cognitive impairment; and progression to dementia. *Neurology* 68: 1790-1789 (2007).
- [119] Anttila T, Helkala EL, Viitamen M, Kåreholt I, Fratiglioni L, Winblad B, *et al.* Alcohol drinking in middle age and subsequent risk of mild cognitive impairment and dementia in old age: a prospective population based study. *BMJ* 329: 539 (2004).
- [120] Espeland MA, Gu L, Masaki KH, Langer RD, Coker LH, Stefanick ML, *et al.* Association between reported alcohol intake and cognition: results from the Women's Health Initiative Memory Study. *Am J Epidemiol* 161: 228-238 (2005).
- [121] Peters R, Peters J, Warner J, Beckett N, Bulpitt C. Alcohol, dementia and cognitive decline in the elderly: a systematic review. *Age Ageing* 37: 505-512 (2008).
- [122] Anstey KJ, Mack HA, Cherbuin N. Alcohol Consumption as a Risk Factor for Dementia and Cognitive Decline: Meta-Analysis of Prospective Studies. *Am J Geriatr Psychiatry* 17: 542-555 (2009).
- [123] Brust JC. A 74-year-old man with memory loss and neuropathy who enjoys alcoholic beverages. *JAMA* 299: 1046-1054 (2008).
- [124] Panza F, Capurso C, Solfrizzi V. Alcohol use, thiamine deficiency, and cognitive impairment. *JAMA* 299: 2853-2854 (2008).
- [125] Hendrie HC, Osuntokun BO, Hall KS, Ogunniyi AO, Hui SL, Unverzagt FW, *et al.* Prevalence of Alzheimer's disease and dementia in two communities: Nigerian Africans and African Americans. *Am J Psychiatry* 152: 1485-1492 (1995).
- [126] White L, Petrovitch H, Ross GW, Masaki KH, Abbott RD, Teng EL, *et al.* Prevalence of dementia in older Japanese-American men in Hawaii. *JAMA* 276: 955-960 (1996).
- [127] Del Parigi A, Panza F, Capurso C, Solfrizzi V. Nutritional factors, cognitive decline, and dementia. *Brain Res Bull* 69: 1-19 (2006).
- [128] Panza F, Capurso C, D'Introno A, Colacicco AM, Del Parigi A, Gagliardi G, *et al.* Mediterranean diet, mild cognitive impairment, and Alzheimer's disease. *Exp Gerontol* 42: 6-7 (2007).
- [129] Luchsinger JA, Tang MX, Shea S, Mayeux R. Caloric intake and the risk of Alzheimer disease. *Arch Neurol* 59: 1256-1263 (2002).
- [130] Steffen LM, Jacobs DR Jr, Stevens J, Shahar E, Carithers T, Folsom AR. Associations of whole-grain, refined-grain, and fruit and vegetable consumption with risks of all-cause mortality and incident coronary artery disease and ischemic stroke: the Atherosclerosis Risk in Communities (ARIC) Study. *Am J Clin Nutr* 78: 383-390 (2003).
- [131] Mizrahi A, Knekt P, Montonen J, Laaksonen MA, Heliövaara M, Järvinen R. Plant foods and the risk of cerebrovascular diseases: a potential protection of fruit consumption. *Br J Nutr* 102: 1075-1083 (2009).
- [132] La Rue A, Koehler KM, Wayne SJ, Chiulli SJ, Haaland KY, Garry PJ. Nutritional status and cognitive functioning in a normally aging sample: a 6-y reassessment. *Am J Clin Nutr* 65: 20-29 (1997).
- [133] Salerno-Kennedy R, Cashman KD. Relationship between dementia and nutrition-related factors and disorders: an overview. *Int J Vit Nutr Res* 75: 83-95 (2005).
- [134] Ancelin ML, Christen Y, Ritchie K. Is antioxidant therapy a viable alternative for mild cognitive impairment? Examination of the evidence. *Dement Geriatr Cogn Disord* 24: 1-19 (2007).
- [135] Grodstein F, Chen J, Willet WC. High-dose antioxidant supplements and cognitive function in community-dwelling elderly women. *Am J Clin Nutr* 77: 3223-3229 (2003).
- [136] Engelhart MJ, Geerlinks MI, Ruitenberg A, van Swieten JC, Hofman A, Witteman JC, *et al.* Dietary intake of antioxidants and risk of Alzheimer disease. *JAMA* 287: 3223-3229 (2002).

- [137] Lim GP, Chu T, Yang F, Beech W, Frautschy SA, Cole GM. The curry spice curcumin reduces oxidative damage and amyloid pathology in an Alzheimer's transgenic mouse. *J Neurosci* 21: 8370-8377 (2001).
- [138] Conte V, Uryu K, Fujimoto S, Yao Y, Rokach J, Longhi L, *et al.* Vitamin E reduces amyloidosis and improves cognitive function in Tg2576 mice following repetitive concussive brain injury. *J Neurochem* 90: 758-764 (2004).
- [139] Huang HY, Caballero B, Chang S, Alberg AJ, Semba RD, Schneyer CR, *et al.* The efficacy and safety of multivitamin and mineral supplement use to prevent cancer and chronic disease in adults: a systematic review for a National Institutes of Health state-of-the-science conference. *Ann Int Med* 145: 372-385 (2006).
- [140] Sano M, Ernesto C, Thomas RG, Klauber MR, Schafer K, Grundman M, *et al.* A controlled trial of selegiline, alpha-tocopherol, or both as treatment for Alzheimer's disease. *N Engl J Med* 336: 1216-1222 (1997).
- [141] Le Bars PL, Velasco VM, Fergusson JM, Dessain EC, Keiser M, Hoerr R. Influence of the severity of cognitive impairment on the effect of Ginkgo biloba extract Egb 761 in Alzheimer's disease. *Neuropsychobiology* 45: 19-26 (2002).
- [142] Petersen RC, Thomas RG, Grundman M, Bennet D, Doody R, Ferris S, *et al.* Vitamin E and donepezil for the treatment of mild cognitive impairment Alzheimer's Disease Cooperative Study Group. *N Engl J Med* 352: 2379-2388 (2005).
- [143] Kamat CD, Gadal S, Mhatre M, Williamson KS, Pye QN, Hensley K. Antioxidants in central nervous system diseases: preclinical promise and translational challenges. *J Alzheimer Dis* 15: 473-493 (2008).
- [144] Lee L, Kang SA, Lee HO, Lee BH, Park JS, Kim JH, *et al.* Relationships between dietary intake and cognitive function level in Korean elderly people. *Public Health* 115: 133-128 (2001).
- [145] Kang JH, Ascherio A, Grodstein F. Fruit and vegetable consumption and cognitive decline in aging women. *Ann Neurol* 57: 713-720 (2005).
- [146] Morris MC, Evans DA, Tangney CC, Bienias JL, Wilson RS. Associations of vegetable and fruit consumption with age-related cognitive change. *Neurology* 67: 1370-1376 (2006).
- [147] Engelhart MJ, Geerlings MI, Ruitenberg A, van Swieten JC, Hofman A, Witteman JC, *et al.* Dietary intake of antioxidants and risk of Alzheimer disease. *JAMA* 287: 3223-3229 (2002).
- [148] Hughes TF, Andel R, Small BJ, Borenstein AR, Mortimer JA, Wolk A, *et al.* Midlife fruit and vegetable consumption and risk of dementia in later life in Swedish twins. *Am J Geriatr Psychiatry* [post author corrections]; DOI: 10.1097/JGP.0b013e3181c65250 (2009).
- [149] Mecocci P, Mariani E, Polidori MC, Hensley K, Butterfield DA. Antioxidant agents in Alzheimer's disease. *Centr Nerv Syst Ag Med Chem* 8: 48-63 (2008).
- [150] Polidori MC, Praticó D, Mangialasche F, Mariani E, Aust O, Anlasik T, *et al.* High fruit and vegetable intake is positively correlated with antioxidant status and cognitive performance in healthy subjects. *J Alzheimers Dis* 17: 921-927 (2009).
- [151] Fratiglioni L, Paillard-Borg S, Winblad B. An active and socially integrated lifestyle in late life might protect against dementia. *Lancet Neurol* 3: 343-353 (2004).
- [152] Féart C, Samieri C, Barberger-Gateau P. Mediterranean diet and cognitive function in older adults. *Curr Opin Clin Nutr Metab Care* 13: 14-18 (2010).
- [153] Kant AK. Dietary patterns and health outcomes. *J Am Diet Assoc* 104: 615-635 (2004).
- [154] Kourlaba G, Panagiotakos DB. Dietary quality indices and human health: a review. *Maturitas* 62: 1-8 (2009).
- [155] Rumawas ME, Dwyer JT, McKeown NM, Meigs JB, Rogers G, Jacques PF. The development of the Mediterranean-style dietary pattern score and its application to the American diet in the Framingham Offspring Cohort. *J Nutr* 139: 1150-1156 (2009).
- [156] Trichopoulos A, Kouris-Blazos A, Wahlqvist ML, Gnardellis C, Lagiou P, Polychronopoulos E, *et al.* Diet and overall survival in elderly people. *BMJ* 311: 1457-1460 (1995).
- [157] Hu FB, Bronner L, Willett WC, Stampfer MJ, Rexrode KM, Albert CM, *et al.* Fish and omega-3 fatty acid intake and risk of coronary heart disease in women. *JAMA* 287: 1815-1821 (2002).
- [158] Scarmeas N, Stern Y, Tang MX, Mayeux R, Luchsinger JA. Mediterranean diet and risk for Alzheimer's disease. *Ann Neurol* 59: 912-921 (2006).
- [159] Scarmeas N, Stern Y, Mayeux R, Luchsinger JA. Mediterranean diet, Alzheimer disease, and vascular mediation. *Arch Neurol* 63: 1709-1717 (2006).
- [160] Scarmeas N, Luchsinger JA, Mayeux R, Stern Y. Mediterranean diet, and Alzheimer disease mortality. *Neurology* 69: 1084-1093 (2007).
- [161] Scarmeas N, Stern Y, Mayeux R, Manly JJ, Schupf N, Luchsinger JA. Mediterranean diet and mild cognitive impairment. *Arch Neurol* 66: 216-225 (2009).
- [162] Féart C, Samieri C, Rondeau V, Amieva H, Portet F, Dartigues JF, *et al.* Adherence to a Mediterranean diet, cognitive decline, and risk of dementia. *JAMA* 302: 638-648 (2009).
- [163] Scarmeas N, Luchsinger JA, Schupf N, Brickman AM, Cosentino S, Tang MX, *et al.* Physical activity, diet, and risk of Alzheimer disease. *JAMA* 302: 627-637 (2009).
- [164] Solfrizzi V, Capurso C, D'Introno A, Colacicco AM, Capurso A, Panza F. Whole-diet approach and risk of chronic disease: Limits and advantages. *J Am Geriatr Soc* 54: 1800-1802 (2006).
- [165] Trichopoulos A, Orfanos P, Norat T, Bueno-de-Mesquita B, Ocké MC, Peeters PH, *et al.* Modified Mediterranean diet and survival: EPIC-elderly prospective cohort study. *BMJ* 330: 1329-1330 (2005).
- [166] Trichopoulos A, Bamia C, Trichopoulos D. Anatomy of health effects of Mediterranean diet: Greek EPIC prospective cohort study. *BMJ* 338: b2337 (2009).
- [167] Solfrizzi V, Frisardi V, Capurso C, D'Introno A, Colacicco AM, Chiloiri R, *et al.* Whole-diet approach: working on a criterion validity for age-related cognitive decline and mild cognitive impairment. *J Am Geriatr Soc* 57: 1944-1946 (2009).
- [168] Jacques PF, Tucker KL. Are dietary patterns useful for understanding the role of diet in chronic disease? *Am J Clin Nutr* 73: 1-2 (2001).
- [169] Renaud S, de Lorgeril M, Delaye J, Jacquard F, Marmelle N, *et al.* Cretan Mediterranean diet for prevention of coronary heart disease. *Am J Clin Nutr* 61: 1360S-1367S (1995).
- [170] Chrysoshoou C, Panagiotakos DB, Pitsavos C, Das UN, Stefanadis C. Adherence to the Mediterranean diet attenuates inflammation and coagulation process in healthy adults: the ATTICA Study. *J Am Coll Cardiol* 44: 152-158 (2004).
- [171] Singh RB, Dubnov G, Niaz MA, Ghosh S, Singh R, Rastogi SS, *et al.* Effect of an Indo-Mediterranean diet on progression of coronary artery disease in high risk patients (Indo-Mediterranean Diet Heart Study): a randomised single-blind trial. *Lancet* 360: 1455-1461 (2002).
- [172] Psaltopoulou T, Naska A, Orfanos P, Trichopoulos D, Moun-takalakis T, Trichopoulos A. Olive oil, the Mediterranean diet, and arterial blood pressure: the Greek European Prospective Investigation into Cancer and Nutrition (EPIC) study. *Am J Clin Nutr* 80: 1012-1018 (2004).
- [173] de Lorgeril M, Salen P, Martin JL, Monjaud I, Delaye J, Marmelle N. Mediterranean diet, traditional risk factors, and the rate of cardiovascular complications after myocardial infarction: final report of the Lyon Diet Heart Study. *Circulation* 99: 779-785 (1999).
- [174] Gauthier S, Reisberg B, Zaudig M, Petersen RC, Ritchie K, Broich K, *et al.* Mild cognitive impairment. *Lancet* 367: 1262-1270 (2006).
- [175] Luchsinger JA, Tang MX, Shea S, Mayeux R. Hyperinsulinemia and risk of Alzheimer disease. *Neurology* 63: 1187-1192 (2004).
- [176] Watson GS, Craft S. The role of insulin resistance in the pathogenesis of Alzheimer's disease: implications for treatment. *CNS Drugs* 17: 27-45 (2003).
- [173] Reger MA, Watson GS, Green PS, Wilkinson CW, Baker LD, Cholerton B, *et al.* Intranasal insulin improves cognition and modulates beta-amyloid in early AD. *Neurology* 70: 440-448 (2008).
- [174] Zhao Z, Xiang Z, Haroutunian V, Buxbaum JD, Stetka B, Pasinetti GM. Insulin degrading enzyme activity selectively decreases in the hippocampal formation of cases at high risk to develop Alzheimer's disease. *Neurobiol Aging* 28: 824-830 (2007).
- [175] Yaffe K, Blackwell T, Kanaya AM, Davidowitz N, Barrett-Connor E, Krueger K. Diabetes, impaired fasting glucose, and development of cognitive impairment in older women. *Neurology* 63: 658-663 (2004).
- [176] Luchsinger JA, Reitz C, Patel B, Tang MX, Manly JJ, Mayeux R. Relation of diabetes to mild cognitive impairment. *Arch Neurol* 64: 570-575 (2007).

- [177] Esposito K, Marfella R, Ciotola M, Di Palo C, Giugliano F, Giugliano G, *et al.* Effect of a Mediterranean-style diet on endothelial dysfunction and markers of vascular inflammation in the metabolic syndrome: a randomized trial. *JAMA* 292: 1440-1446 (2004).
- [178] Jula A, Marniemi J, Huupponen R, Virtanen A, Rastas M, Rönnemaa T. Effects of diet and simvastatin on serum lipids, insulin, and antioxidants in hypercholesterolemic men: a randomized controlled trial. *JAMA* 287: 598-605 (2002).
- [179] Dai J, Jones DP, Goldberg J, Bostick RM, Wilson PW, Manatunga AK, *et al.* Association between adherence to the Mediterranean diet and oxidative stress. *Am J Clin Nutr* 88: 1364-1370 (2008).
- [180] Harris MI, Flegal KM, Cowie CC, Eberhardt MS, Goldstein DE, Little RR, *et al.* Prevalence of diabetes, impaired fasting glucose, and impaired glucose tolerance in U.S. adults: the Third National Health and Nutrition Examination Survey, 1988-1994. *Diabetes Care* 21: 518-524 (1998).
- [181] Giasson BI, Ischiropoulos H, Lee VM, Trojanowski JQ. The relationship between oxidative/nitritative stress and pathological inclusions in Alzheimer's and Parkinson's diseases. *Free Radic Biol Med* 32: 1264-1275 (2002).
- [182] Ding Q, Dimayuga E, Keller JN. Oxidative damage, protein synthesis, and protein degradation in Alzheimer's disease. *Curr Alzheimer Res* 4: 73-79 (2007).
- [183] Keller JN, Schmitt FA, Scheff SW, Ding Q, Chen Q, Butterfield DA, *et al.* Evidence of increased oxidative damage in subjects with mild cognitive impairment. *Neurology* 64: 1152-1156 (2005).
- [184] Mecocci P. Oxidative stress in mild cognitive impairment and Alzheimer disease: a continuum. *J Alzheimers Dis* 6: 159-163 (2004).
- [185] Fito M, Cladellas M, de la Torre R, Martí J, Alcántara M, Pujadas-Bastardes M, *et al.* Antioxidant effect of virgin olive oil in patients with stable coronary heart disease: a randomized, crossover, controlled, clinical trial. *Atherosclerosis* 181: 149-158 (2005).
- [186] Szeto YT, Tomlinson B, Benzie IF. Total antioxidant and ascorbic acid content of fresh fruits and vegetables: implications for dietary planning and food preservation. *Br J Nutr* 87: 55-59 (2002).
- [187] Byers T, Perry G. Dietary carotenes, vitamin C, and vitamin E as protective antioxidants in human cancers. *Annu Rev Nutr* 12: 139-159 (1992).
- [188] Stupans I, Kirlich A, Tuck KL, Hayball PJ. Comparison of radical scavenging effect, inhibition of microsomal oxygen free radical generation, and serum lipoprotein oxidation of several natural antioxidants. *J Agric Food Chem* 50: 2464-2469 (2002).
- [189] Mancini M, Parfitt VJ, Rubba P. Antioxidants in the Mediterranean diet. *Can J Cardiol* 11(suppl G): 105G-109G (1995).
- [190] Blum S, Aviram M, Ben-Amotz A, Levy Y. Effect of a Mediterranean meal on postprandial carotenoids, paraoxonase activity and C-reactive protein levels. *Ann Nutr Metab* 50: 20-24 (2006).
- [191] Wallace AJ, Sutherland WH, Mann JI, Williams SM. The effect of meals rich in thermally stressed olive and safflower oils on postprandial serum paraoxonase activity in patients with diabetes. *Eur J Clin Nutr* 55: 951-958 (2001).
- [192] Aviram M, Dornfeld L, Rosenblat M, Volkova N, Kaplan M, Coleman R, *et al.* Pomegranate juice consumption reduces oxidative stress, atherogenic modifications to LDL, and platelet aggregation: studies in humans and in atherosclerotic apolipoprotein E-deficient mice. *Am J Clin Nutr* 71: 1062-1076 (2000).
- [193] Sanchez-Moreno C, Cano MP, de Ancos B, Plaza L, Olmedilla B, Granado F, *et al.* Mediterranean vegetable soup consumption increases plasma vitamin C and decreases F2-isoprostanes, prostaglandin E2 and monocyte chemotactic protein-1 in healthy humans. *J Nutr Biochem* 17: 183-189 (2006).
- [194] Giugliano D, Esposito K. Mediterranean diet and metabolic diseases. *Curr Opin Lipidol* 19: 63-68 (2008).
- [195] Panagiotakos DB, Dimakopoulou K, Katsouyanni K, Bellander T, Grau M, Koenig W, *et al.* Mediterranean diet and inflammatory response in myocardial infarction survivors. *Int J Epidemiol* 38: 856-866 (2009).
- [196] Wenering SC, Yankner BA. Inflammation and Alzheimer disease: the good, the bad, and the ugly. *Nat Med* 7: 527-528 (2001).
- [197] Walsh S, Aisen PS. Inflammatory processes and Alzheimer's disease. *Expert Rev Neurother* 4: 793-798 (2004).
- [198] Galimberti D, Schoonenboom N, Scheltens P, Fenoglio C, Bouwman F, Venturini E, *et al.* Intrathecal chemokine synthesis in mild cognitive impairment and Alzheimer disease. *Arch Neurol* 63: 538-543 (2006).
- [199] Guerreiro RJ, Santana I, Bras JM, Santiago B, Paiva A, Oliveira C. Peripheral inflammatory cytokines as biomarkers in Alzheimer's disease and mild cognitive impairment. *Neurodegener Dis* 4: 406-412 (2007).
- [200] Magaki S, Mueller C, Dickson C, Kirsch W. Increased production of inflammatory cytokines in mild cognitive impairment. *Exp Gerontol* 42: 233-240 (2007).
- [201] Iwamoto N, Nishiyama E, Ohwada J, Arai H. Demonstration of CRP immunoreactivity in brains of Alzheimer's disease: immunohistochemical study using formic acid pretreatment of tissue sections. *Neurosci Lett* 177: 23-26 (1994).
- [202] Duong T, Nikolaeva M, Acton PJ. C-reactive protein-like immunoreactivity in the neurofibrillary tangles of Alzheimer's disease. *Brain Res* 749: 152-156 (1997).
- [203] McGeer EG, Yasojima K, Schwab C, McGeer PL. The pentraxins: possible role in Alzheimer's disease and other innate inflammatory diseases. *Neurobiol Aging* 22: 843-848 (2001).
- [204] Wood JA, Wood PL, Ryan R, Graff-Radford NR, Pilapil C, Robitaille Y, *et al.* Cytokine indices in Alzheimer's temporal cortex: no changes in mature IL-1 beta or IL-1RA but increases in the associated acute phase proteins IL-6, alpha 2-macroglobulin and C-reactive protein. *Brain Res* 629: 245-252 (1993).
- [205] Schmidt R, Schmidt H, Curb JD, Masaki K, White LR, Launer LJ. Early inflammation and dementia: a 25-year follow-up of the Honolulu-Asia Aging Study. *Ann Neurol* 52: 168-174 (2002).
- [206] Frank RA, Galasko D, Hampel H, Hardy J, de Leon MJ, Mehta PD, *et al.* Biological markers for therapeutic trials in Alzheimer's disease: proceedings of the biological markers working group; NIA initiative on neuroimaging in Alzheimer's disease. *Neurobiol Aging* 24: 521-536 (2003).
- [207] Roberts RO, Geda YE, Knopman DS, Boeve BF, Christianson TJ, Pankratz VS, *et al.* Association of C-reactive protein with mild cognitive impairment. *Alzheimers Dement* 5: 398-405 (2009).
- [208] Schuitmaker A, Dik MG, Veerhuis R, Scheltens P, Schoonenboom NS, Hack CE, *et al.* Inflammatory markers in AD and MCI patients with different biomarker profiles. *Neurobiol Aging* 30, 1885-1889 (2009).
- [209] Paschos GK, Rallidis LS, Liakos GK, Panagiotakos D, Anastasiadis G, Votvas V, *et al.* Background diet influences the anti-inflammatory effect of alpha-linolenic acid in dyslipidaemic subjects. *Br J Nutr* 92: 649-655 (2004).
- [210] Fung TT, McCullough ML, Newby P, *et al.* Diet-quality scores and plasma concentrations of markers of inflammation and endothelial dysfunction. *Am J Clin Nutr* 82: 163-173 (2005).
- [211] Weaver JD, Huang MH, Albert M, Harris T, Rowe JW, Seeman TE. Interleukin-6 and risk of cognitive decline: MacArthur studies of successful aging. *Neurology* 59: 371-378 (2002).
- [212] Engelhart MJ, Geerlings MI, Meijer J, Kilian A, Ruitenberg A, van Swieten JC, *et al.* Inflammatory proteins in plasma and the risk of dementia: the Rotterdam study. *Arch Neurol* 61: 668-672 (2004).
- [213] Bertelli AA, Migliori M, Panichi V, Longoni B, Origlia N, Ferretti A, *et al.* Oxidative stress and inflammatory reaction modulation by white wine. *Ann N Y Acad Sci* 957: 295-301 (2002).
- [214] Greenwood CE, Winocur G. Cognitive impairment in rats fed high-fat diets: a specific effect of saturated fatty acid intake. *Behav Neurosci* 110: 451-459 (1996).
- [215] Moriguchi T, Greiner RS, Salem N. Behavioral deficits associated with dietary induction of decreased brain docosahexaenoic acid concentration. *J Neurochem* 75: 2563-2573 (2000).
- [216] Hashimoto M, Tanabe Y, Fujii Y, Kikuta T, Shibata H, Shido O. Chronic administration of docosahexaenoic acid ameliorates the impairment of spatial cognition learning ability in amyloid beta-infused rats. *J Nutr* 135: 549-555 (2005).
- [217] Hashimoto M, Hossain S, Shimada T, Sugioka K, Yamasaki H, Fujii Y, *et al.* Docosahexaenoic acid provides protection from impairment of learning ability in Alzheimer's disease model rats. *J Neurochem* 81: 1084-1091 (2002).
- [218] Calon F, Lim GP, Yang F, Morihara T, Teter B, Ubeda O, *et al.* Docosahexaenoic acid protects from dendritic pathology in an Alzheimer's disease mouse model. *Neuron* 43: 633-645 (2004).
- [219] Florent S, Malaplate-Armand C, Youssef I, Kriem B, Koziel V, Escanye M-C, *et al.* Docosahexaenoic acid prevents neuronal apoptosis induced by soluble amyloid oligomers. *J Neurochem* 96: 385-395 (2006).

- [220] Cole GM, Lim GP, Yang F, Teter B, Begum A, Ma Q, *et al.* Prevention of Alzheimer's disease: Omega-3 fatty acid and phenolic anti-oxidant interventions. *Neurobiol Aging* 26 (Suppl 1): 133–136 (2005).
- [221] Green KN, Martinez-Coria H, Khashwji H, Hall EB, Yurko-Mauro KA, Ellis L, *et al.* Dietary docosahexaenoic acid and docosapentaenoic acid ameliorate amyloid-beta and tau pathology via a mechanism involving presenilin 1 levels. *J Neurosci* 27: 4385–4395 (2007).
- [222] Park IH, Hwang EM, Hong HS, Boo JH, Oh SS, Lee J, *et al.* Lovastatin enhances Abeta production and senile plaque deposition in female Tg2576 mice. *Neurobiol Aging* 24: 637–643 (2003).
- [223] Lim GP, Calon F, Moriguchi T, Yang F, Teter B, Ubeda O, *et al.* A diet enriched with the omega-3 fatty acid docosahexaenoic acid reduces amyloid burden in an aged Alzheimer mouse model. *J Neurosci* 25: 3032–3040 (2005).
- [224] Oksman M, Iivonen H, Högberg E, Amtul Z, Penke B, Leenders I, *et al.* Impact of different saturated fatty acid, polyunsaturated fatty acid and cholesterol containing diets on beta-amyloid accumulation in APP/PS1 transgenic mice. *Neurobiol Dis* 23: 563–572 (2006).
- [225] Molteni R, Barnard RJ, Ying Z, Roberts CK, Gómez-Pinilla F. A high-fat, refined sugar diet reduces hippocampal brain-derived neurotrophic factor, neuronal plasticity, and learning. *Neuroscience* 112: 803–814 (2002).
- [226] Ambring A, Friberg P, Axelsen M, Laffrenzen M, Taskinen MR, Basu S, *et al.* Effects of a Mediterranean-inspired diet on blood lipids; vascular function and oxidative stress in healthy subjects. *Clin Sci* 106: 519–525 (2004).
- [227] Scientific Advisory Committee on Nutrition and the Committee on Toxicity. Advice on fish consumption: benefits and risks. London: The Stationery Office (2004).
- [228] Annweiler C, Allali G, Allain P, Bridenbaugh S, Schott AM, Kressig RW, *et al.* Vitamin D and cognitive performance in adults: a systematic review. *Eur J Neurol* 16: 1083–1089 (2009).
- [229] Annweiler C, Schott AM, Allali G, Bridenbaugh SA, Kressig RW, Allain P, *et al.* Association of vitamin D deficiency with cognitive impairment in older women. Cross-sectional study. *Neurology* 74: 27–32 (2010).
- [230] Mukamal KJ, Longstreth WT Jr, Mittleman MA, Crum RM, Siscovick DS. Alcohol consumption and subclinical findings on magnetic resonance imaging of the brain in older adults: the cardiovascular health study. *Stroke* 32: 1939–1946 (2001).
- [231] Fadda F, Rossetti ZL. Chronic ethanol consumption: from neuroadaptation to neurodegeneration. *Prog Neurobiol* 56: 385–431 (1998).
- [232] National Institute on Alcohol Abuse and Alcoholism. Alcohol Alert No. 46: Are Women More Vulnerable to Alcohol's Effects? Bethesda, MD: National Institutes of Health (1999).
- [233] Stancampiano R, Carta M, Cocco S, Curreli R, Rossetti ZL, Fadda F. Biphasic effects of ethanol on acetylcholine release in the rat prefrontal cortex. *Brain Res* 997: 128–132 (2004).
- [234] Naimi TS, Brown DW, Brewer RD, Giles WH, Mensah G, Serdula MK, *et al.* Cardiovascular risk factors and confounders among nondrinking and moderate-drinking US adults. *Am J Prev Med* 28: 369–373 (2005).
- [235] Palomaki H, Kaste M. Regular light-to-moderate intake of alcohol and the risk of ischemic stroke: is there a beneficial effect? *Stroke* 24: 1828–1832 (1993).
- [236] den Heijer T, Vermeer SE, van Dijk EJ, Prins ND, Koudstaal PJ, van Duijn CM, *et al.* Alcohol intake in relation to brain magnetic resonance imaging findings in older persons without dementia. *Am J Clin Nutr* 80: 992–997 (2004).
- [237] Mukamal KJ, Kuller LH, Fitzpatrick AL, Longstreth WT Jr, Mittleman MA, Siscovick DS. Prospective study of alcohol consumption and risk of dementia in older adults. *JAMA* 289: 1405–1413 (2003).
- [238] Serafini M, Maiani G, Ferro-Luzzi A. Alcohol-free red wine enhances plasma antioxidant capacity in humans. *J Nutr* 128: 1003–1007 (1998).
- [239] Waterhouse AL. Wine phenolics. *Ann N Y Acad Sci* 957: 21–36 (2002).
- [240] Commenges D, Scotet V, Renaud S, Jacqmin-Gadda H, Barberger-Gateau P, Dartigues J-F. Intake of flavonoids and risk of dementia. *Eur J Epidemiol* 16: 357–363 (2000).
- [241] Heinonen IM, Lehtonen PJ, Hopia AI. Antioxidant activity of berry and fruit wines and liquors. *J Agric Food Chem* 46: 25–31 (1998).
- [242] Albani D, Polito L, Forloni G. Sirtuins as novel targets for Alzheimer's disease and other neurodegenerative disorders: experimental and genetic evidence. *J Alzheimers Dis* 19: 11–26 (2010).
- [243] Wenzel E, Somoza V. Metabolism and bioavailability of trans-resveratrol. *Mol Nutr Food Res* 49: 472–481 (2005).
- [244] Walle T, Hsieh F, DeLegge MH, Oatis JE, Walle UK. High absorption but very low bioavailability of oral resveratrol in humans. *Drug Metab Dispos* 32: 1377–1382 (2004).
- [245] Van Dyk K, Sano M. The impact of nutrition on cognition in the elderly. *Neurochem Res* 32: 893–904 (2007).
- [246] Letenneur L, Proust-Lima C, Le Gouge A, Dartigues J, Barberger-Gateau P. Flavonoid intake and cognitive decline over a 10-year period. *Am J Epidemiol* 165: 1364–1371 (2007).
- [247] Smith MA, Hirai K, Hsiao K, Pappolla MA, Harris PL, Siedlak SL, *et al.* Amyloid-beta deposition in Alzheimer transgenic mice is associated with oxidative stress. *J Neurochem* 70: 2212–2215 (1998).
- [248] Youdim KA, Kaiser MZ, Begley DJ, Rice-Evans CA, Abbott NJ. Flavonoid permeability across an in situ model of the blood-brain barrier. *Free Radic Biol Med* 36: 592–604 (2004).
- [249] Ortiz D, Shea TB. Apple juice prevents oxidative stress induced by amyloid-beta in culture. *J Alzheimers Dis* 6: 27–30 (2004).
- [250] Heo HJ, Lee CY. Protective effects of quercetin and vitamin C against oxidative stress-induced neurodegeneration. *J Agric Food Chem* 52: 7514–7517 (2004).
- [251] Nakayama T. Suppression of hydroperoxide-induced cytotoxicity by polyphenols. *Cancer Res* 54(7 Suppl): 1991s–1993s (1994).
- [252] Patil CS, Singh VP, Satyanarayan PS, Jain NK, Singh A, Kulkarni SK. Protective effect of flavonoids against aging- and lipopolysaccharide-induced cognitive impairment in mice. *Pharmacology* 69: 59–67 (2003).
- [253] Rogers EJ, Milhalik S, Orthiz D, Shea TB. Apple juice prevents oxidative stress and impaired cognitive performance caused by genetic and dietary deficiencies in mice. *J Nutr Health Aging* 8: 92–97 (2004).
- [254] Morris MC, Evans DA, Bienias JL, Tangney CC, Wilson RS. Vitamin E and cognitive decline in older persons. *Arch Neurol* 59: 1125–1132 (2002).
- [255] Dauchet L, Amouyel P, Dallongeville J. Fruit and vegetable consumption and risk of stroke: a meta-analysis of cohort studies. *Neurology* 65: 1193–1197 (2005).
- [256] Wannamethee SG, Lowe GD, Rumley A, Bruckdorfer KR, Whincup PH. Associations of vitamin C status, fruit and vegetable intakes, and markers of inflammation and hemostasis. *Am J Clin Nutr* 83: 567–574 (2006).
- [257] Manach C, Scalbert A, Morand C, Remesy C, Jimenez L. Polyphenols: food sources and bioavailability. *Am J Clin Nutr* 79: 727–747 (2004).
- [258] Sanchez-Moreno C, Cano MP, de Ancos B, Plaza L, Olmedilla B, Granado F, *et al.* High-pressurized orange juice consumption affects plasma vitamin C, antioxidative status and inflammatory markers in healthy humans. *J Nutr* 133: 2204–2209 (2003).
- [259] Kiefer I, Prock P, Lawrence C, Wise J, Bieger W, Bayer P, *et al.* Supplementation with mixed fruit and vegetable juice concentrates increased serum antioxidants and folate in healthy adults. *J Am Coll Nutr* 23: 205–211 (2004).
- [260] Wang HX, Wahlin A, Basun H, Fastbom J, Winblad B, Fratiglioni L. Vitamin B(12) and folate in relation to the development of Alzheimer's disease. *Neurology* 56: 1188–1194 (2001).